

BFS Traversal

Step 1: Start at A

Legend

Complete

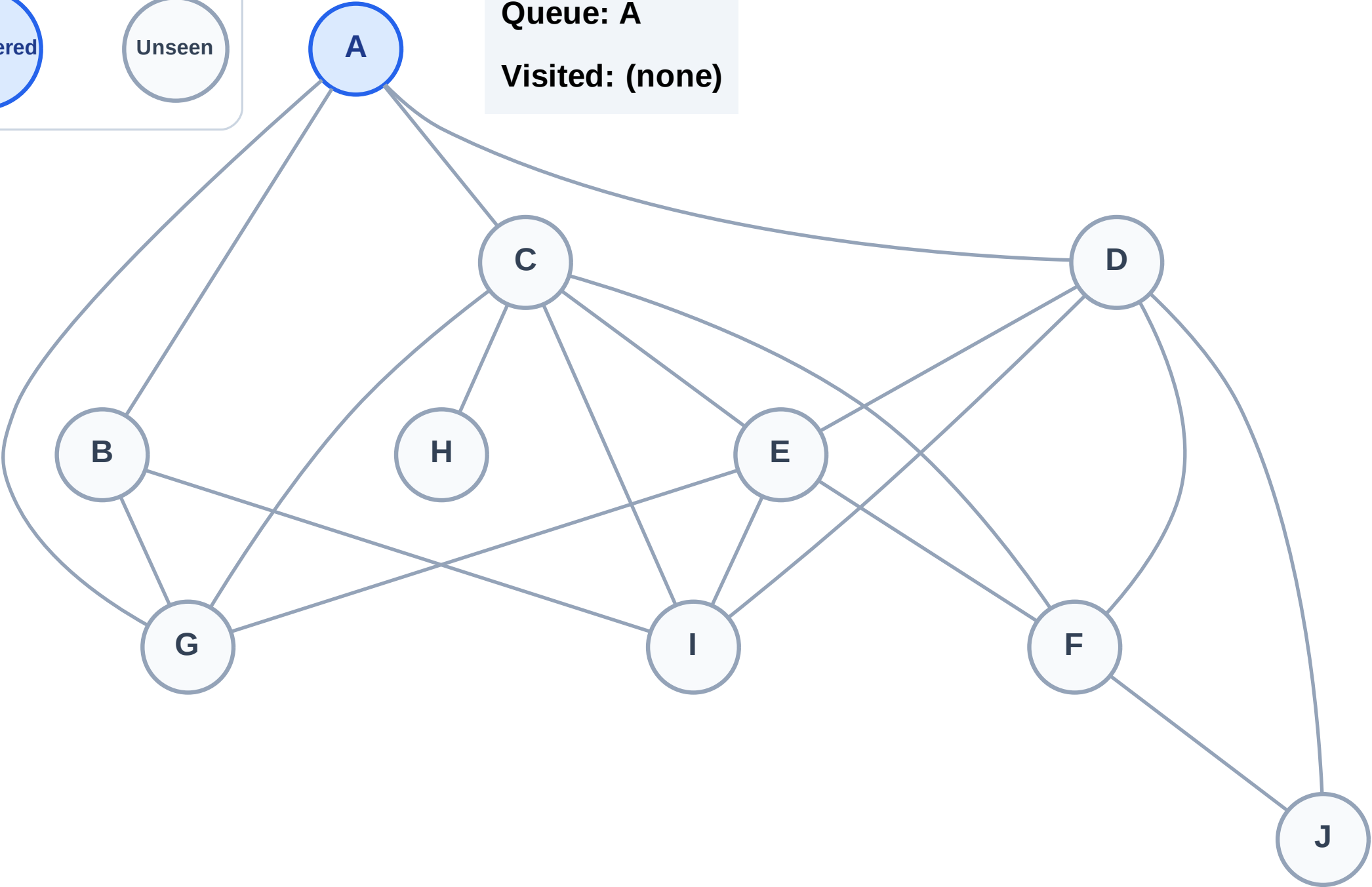
Processing

Discovered

Unseen

Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Complete

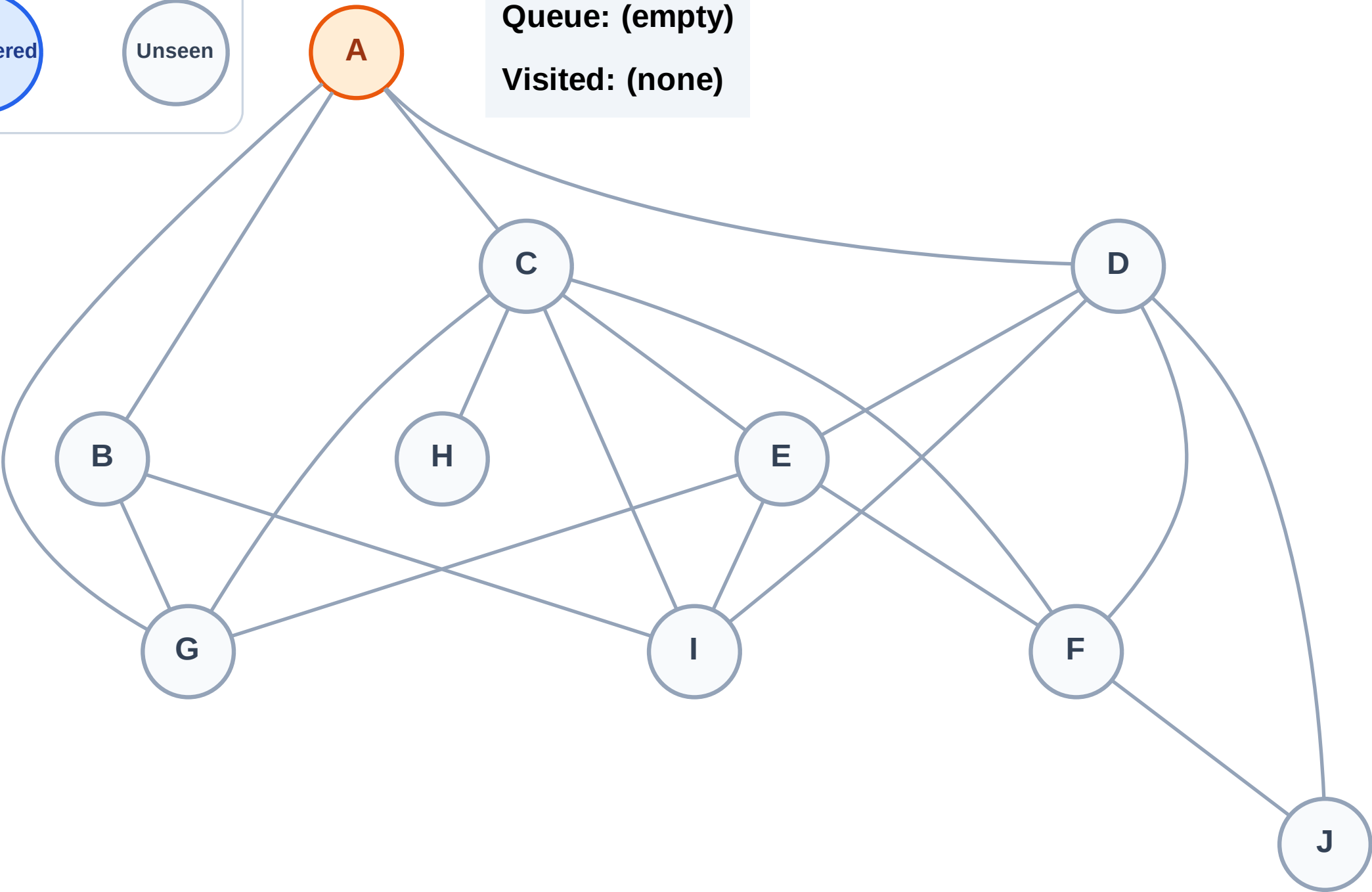
Processing

Discovered

Unseen

Queue: (empty)

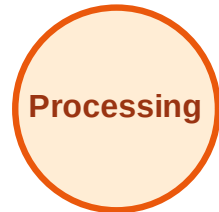
Visited: (none)



BFS Traversal

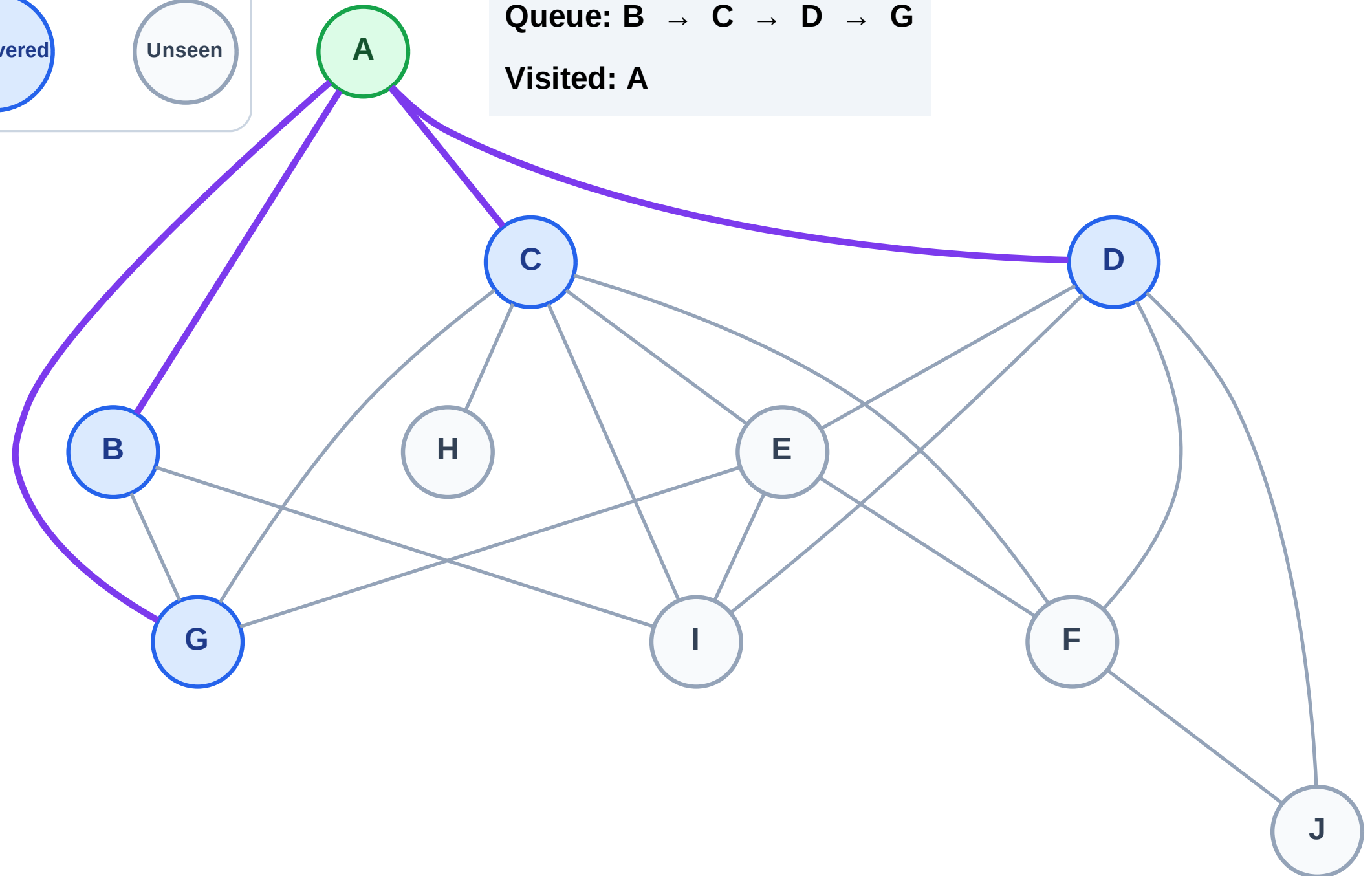
Step 3: Discover B, C, D, G from A

Legend



Queue: B → C → D → G

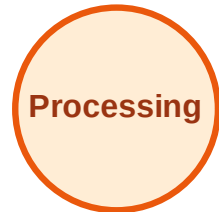
Visited: A



BFS Traversal

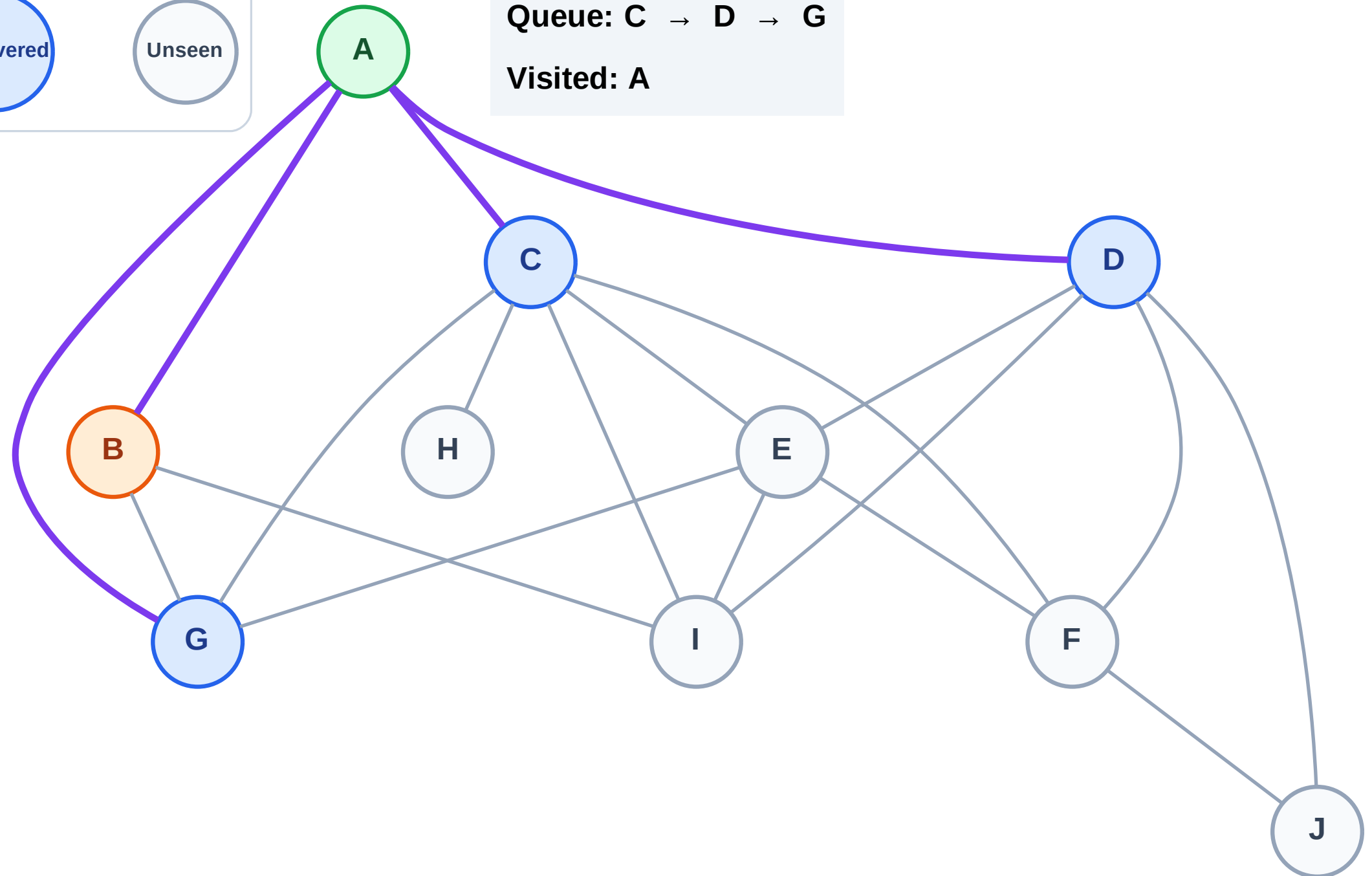
Step 4: Process node B

Legend



Queue: C → D → G

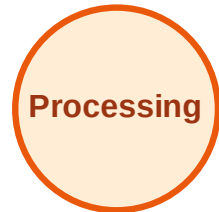
Visited: A



BFS Traversal

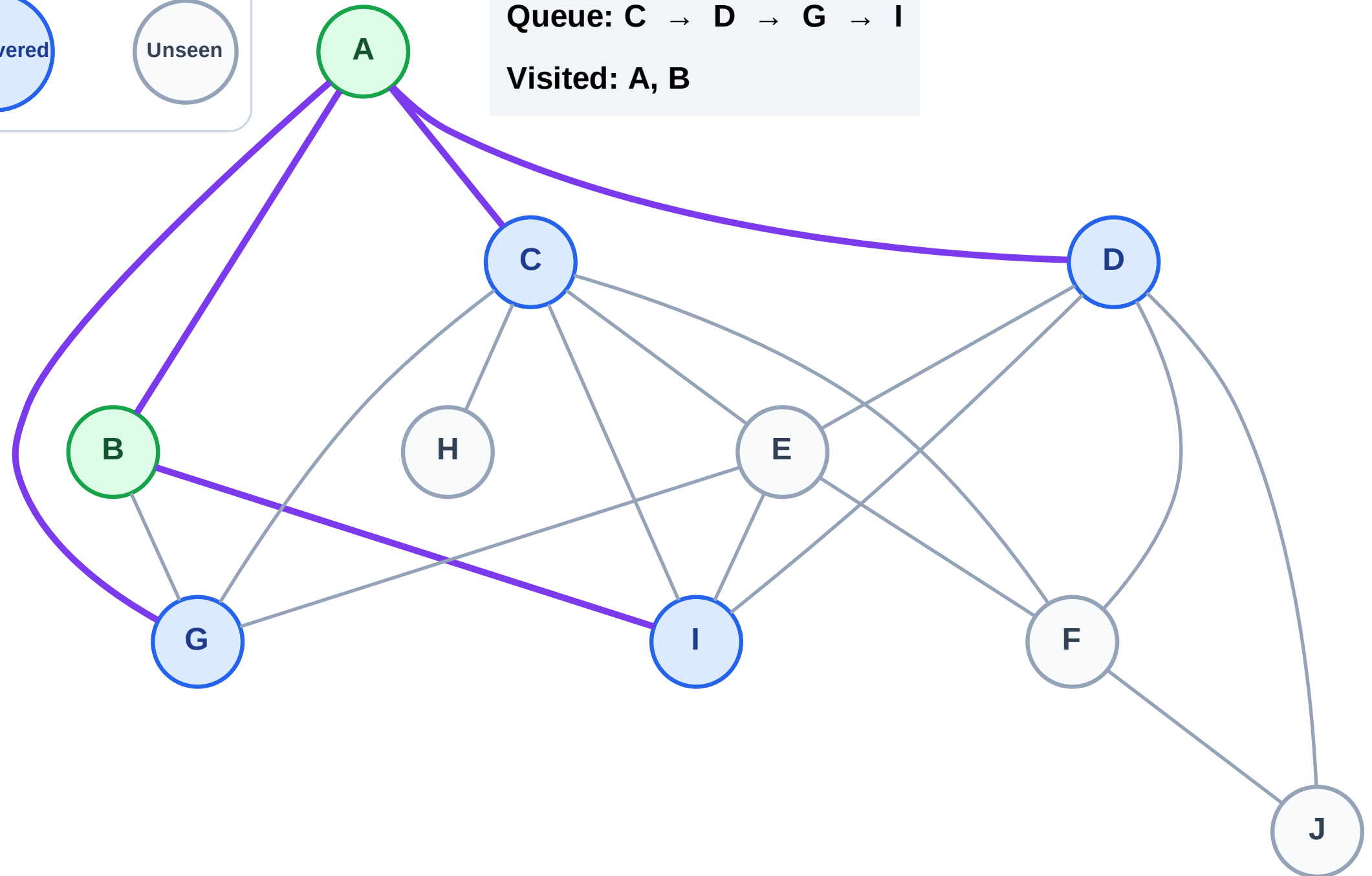
Step 5: Discover I from B

Legend



Queue: C → D → G → I

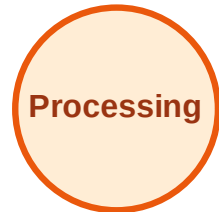
Visited: A, B



BFS Traversal

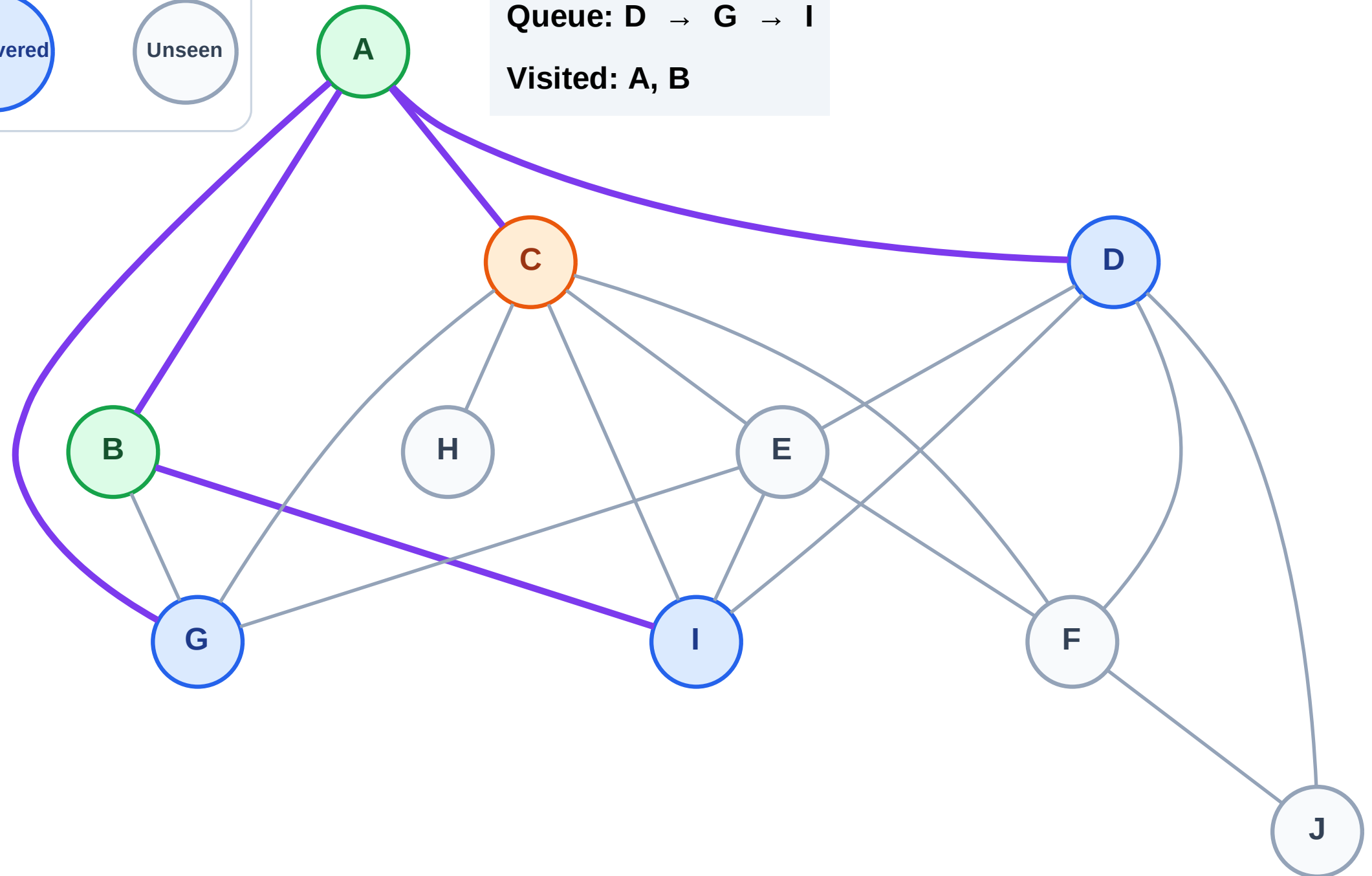
Step 6: Process node C

Legend



Queue: D → G → I

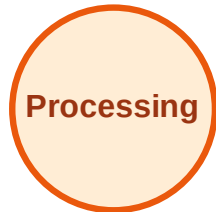
Visited: A, B



BFS Traversal

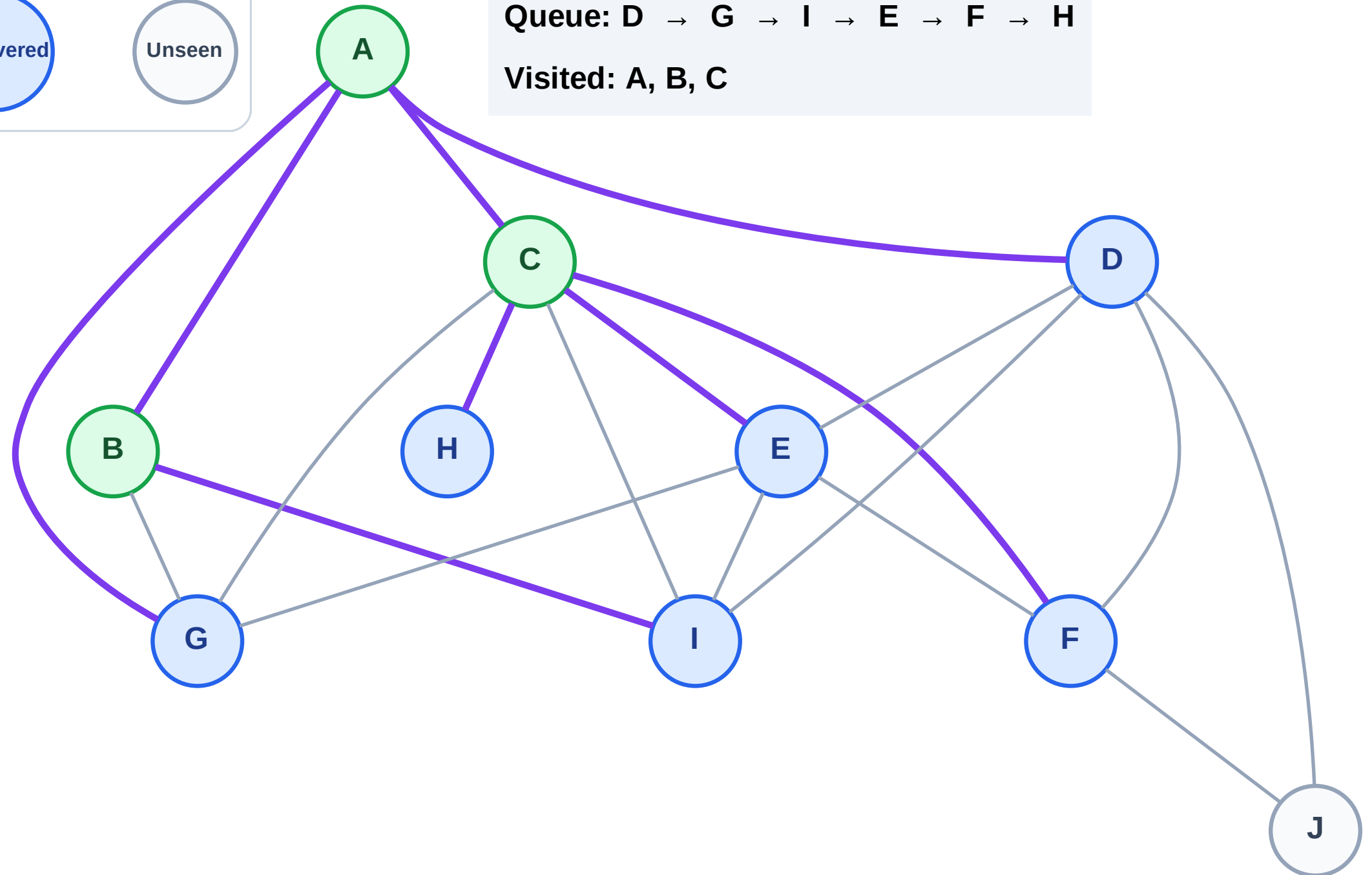
Step 7: Discover E, F, H from C

Legend



Queue: D → G → I → E → F → H

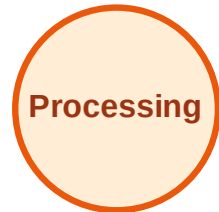
Visited: A, B, C



BFS Traversal

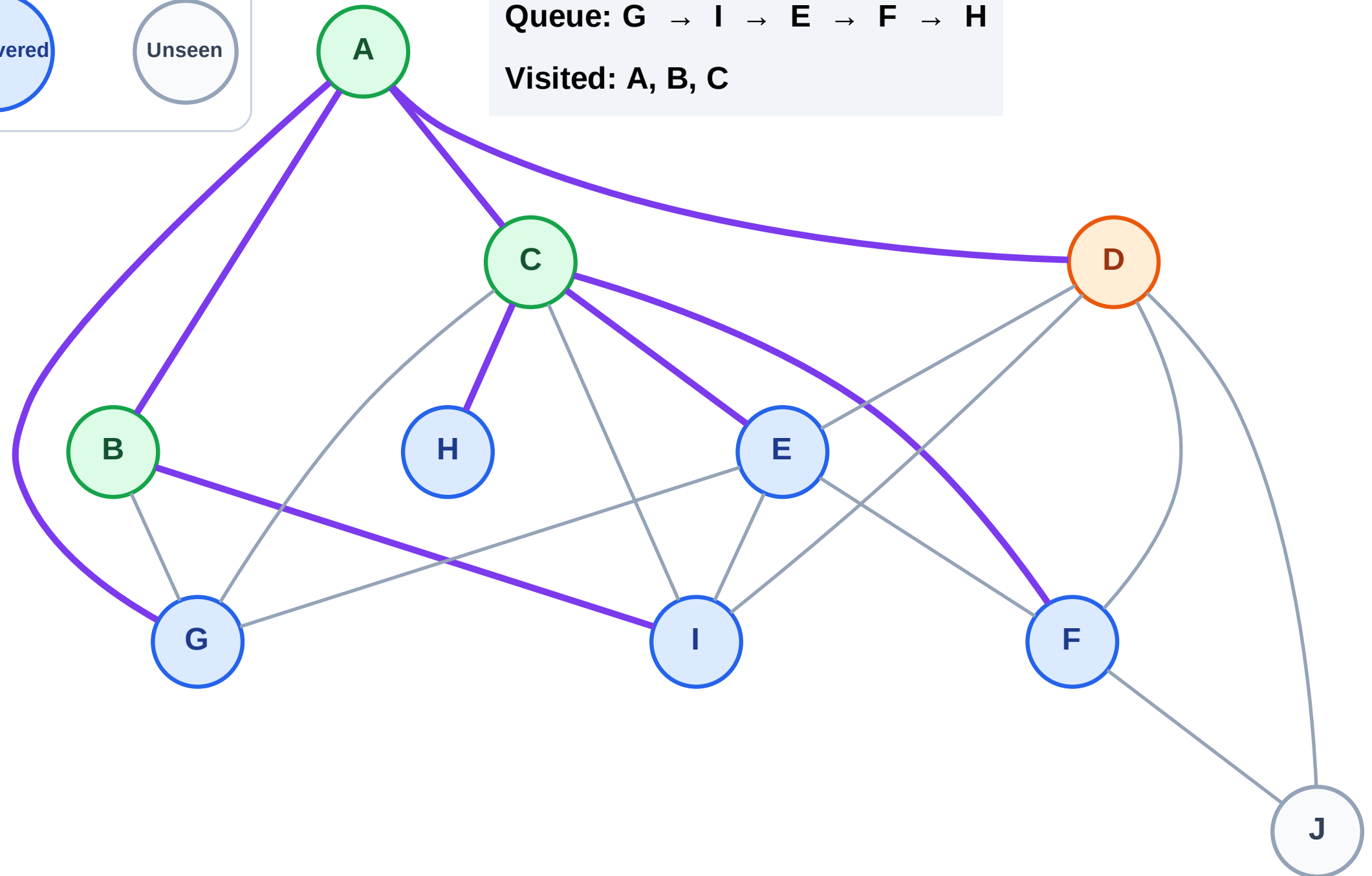
Step 8: Process node D

Legend



Queue: G → I → E → F → H

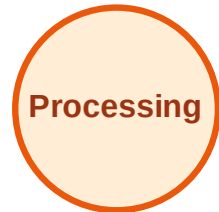
Visited: A, B, C



BFS Traversal

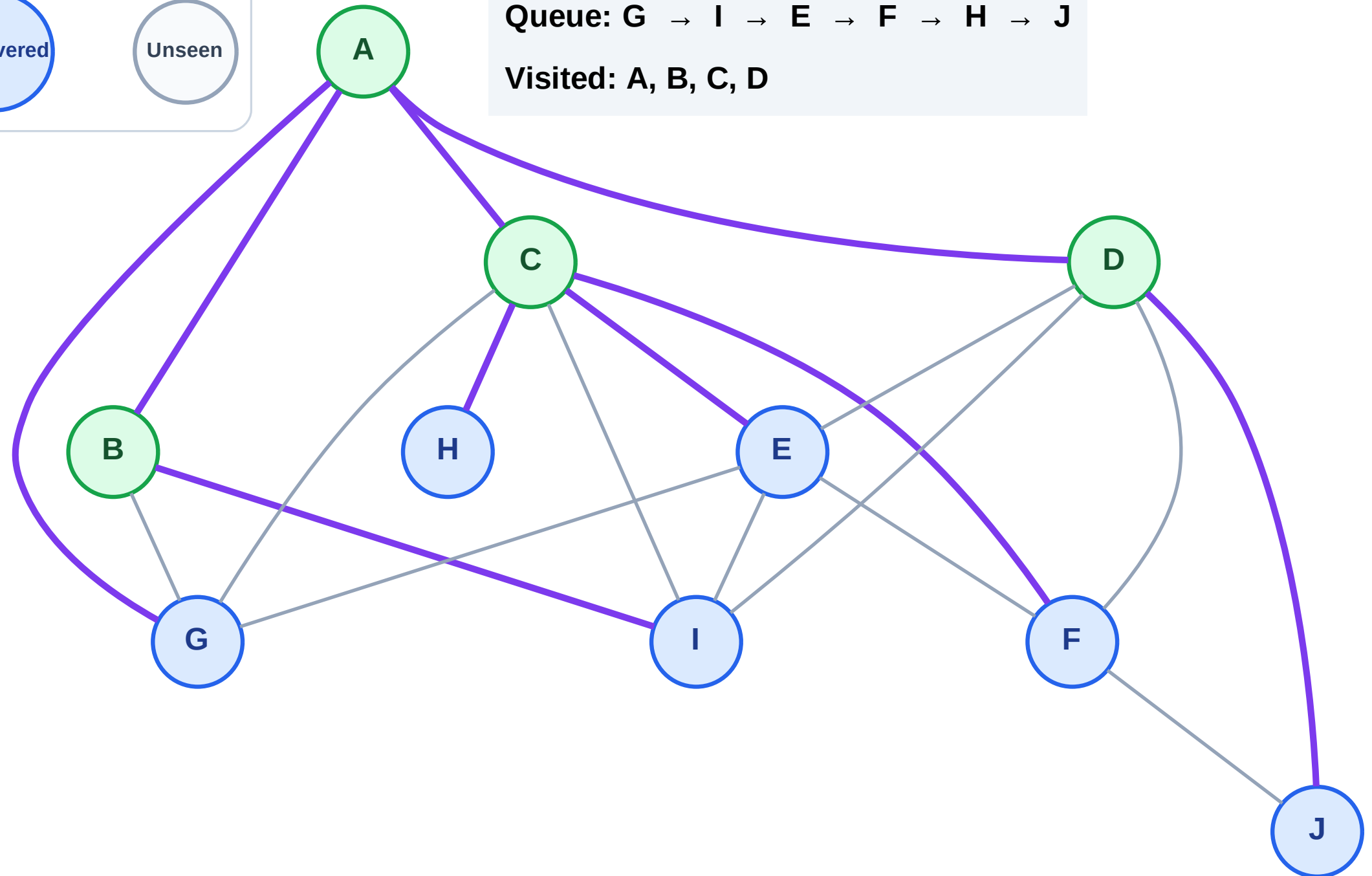
Step 9: Discover J from D

Legend



Queue: G → I → E → F → H → J

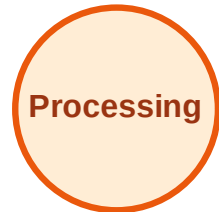
Visited: A, B, C, D



BFS Traversal

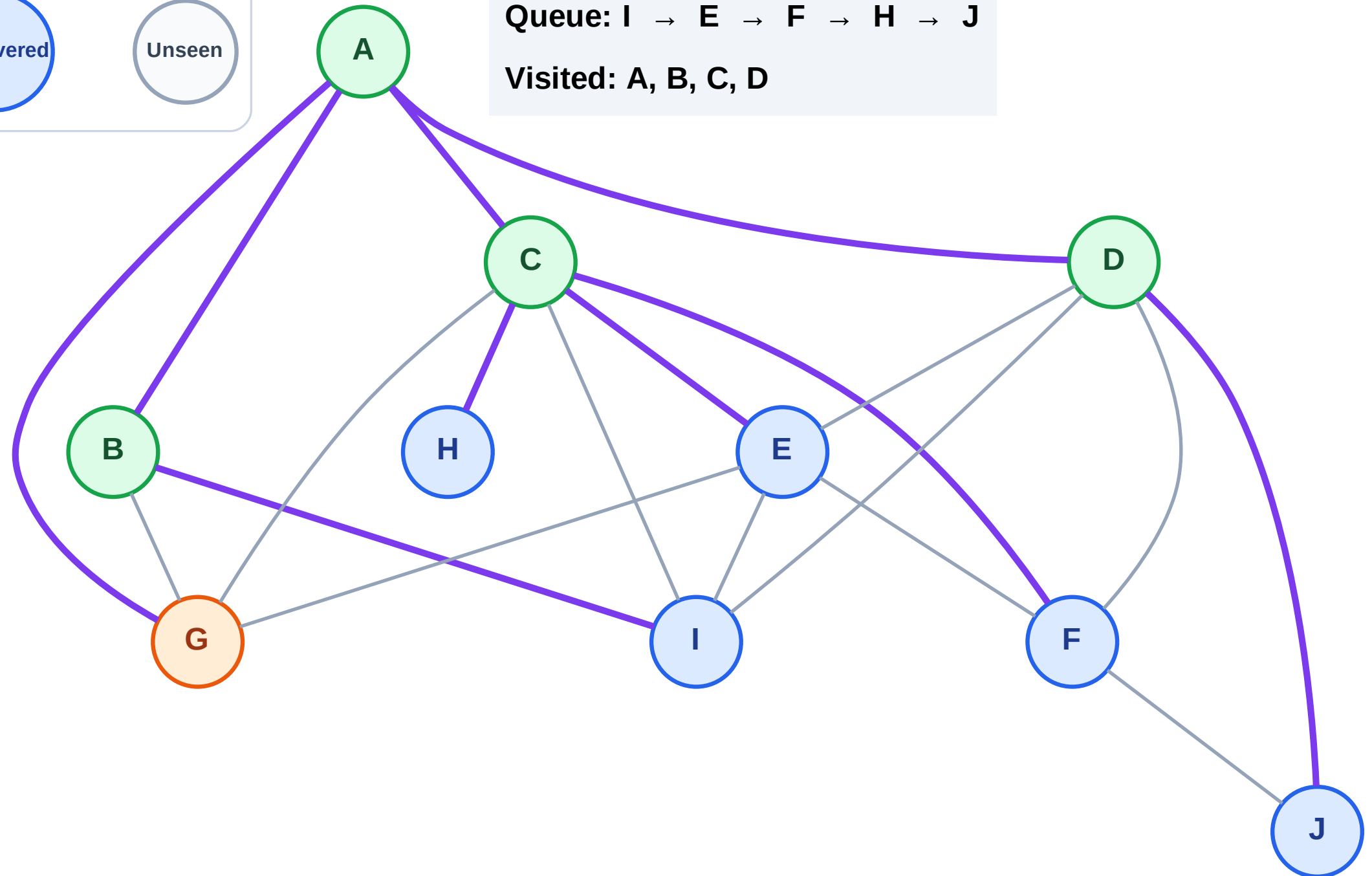
Step 10: Process node G

Legend



Queue: I → E → F → H → J

Visited: A, B, C, D



BFS Traversal

Step 11: No new neighbors from G

Legend

Complete

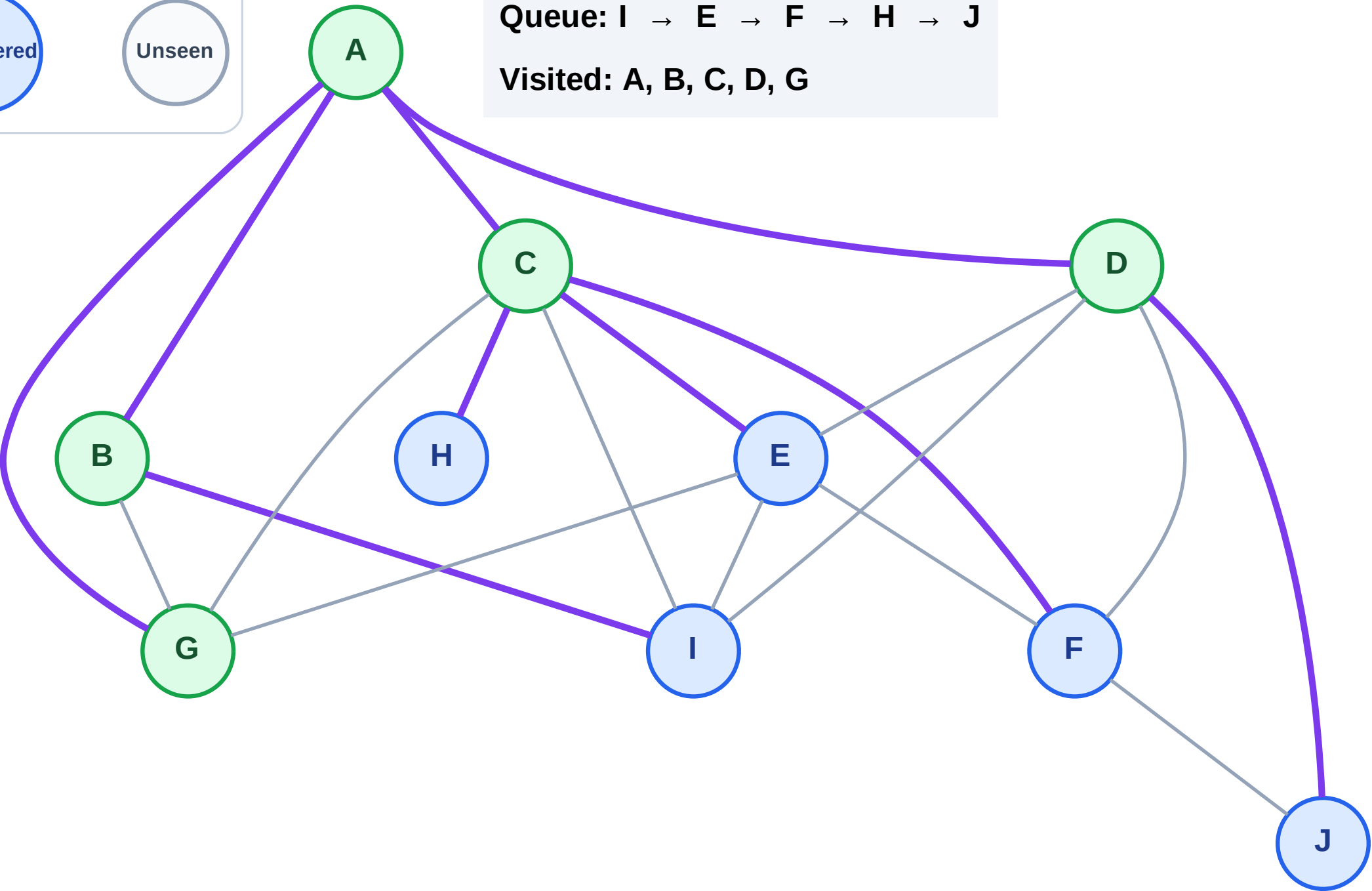
Processing

Discovered

Unseen

Queue: I → E → F → H → J

Visited: A, B, C, D, G



BFS Traversal

Step 12: Process node I

Legend

Complete

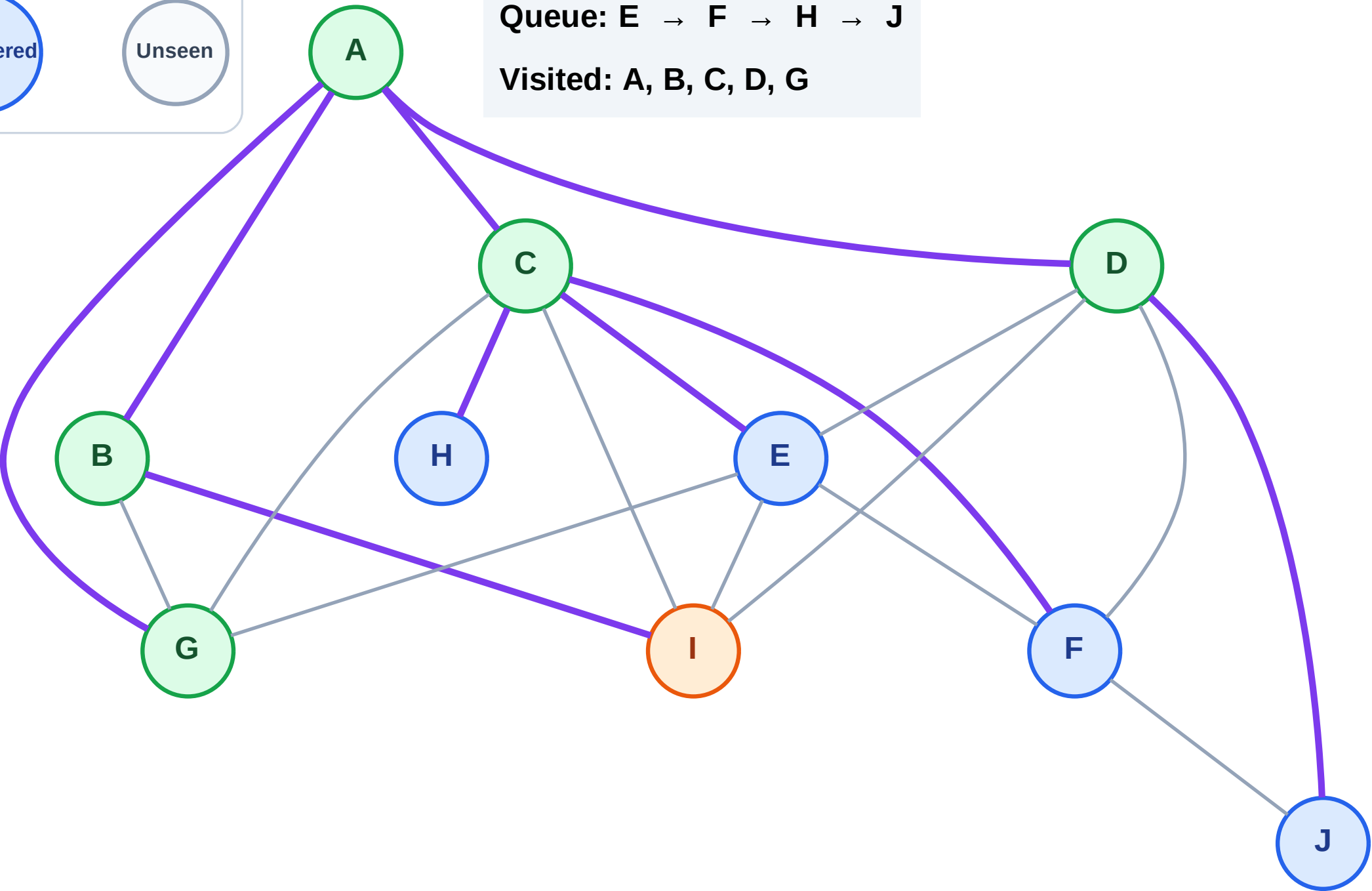
Processing

Discovered

Unseen

Queue: E → F → H → J

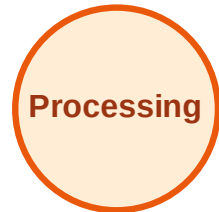
Visited: A, B, C, D, G



BFS Traversal

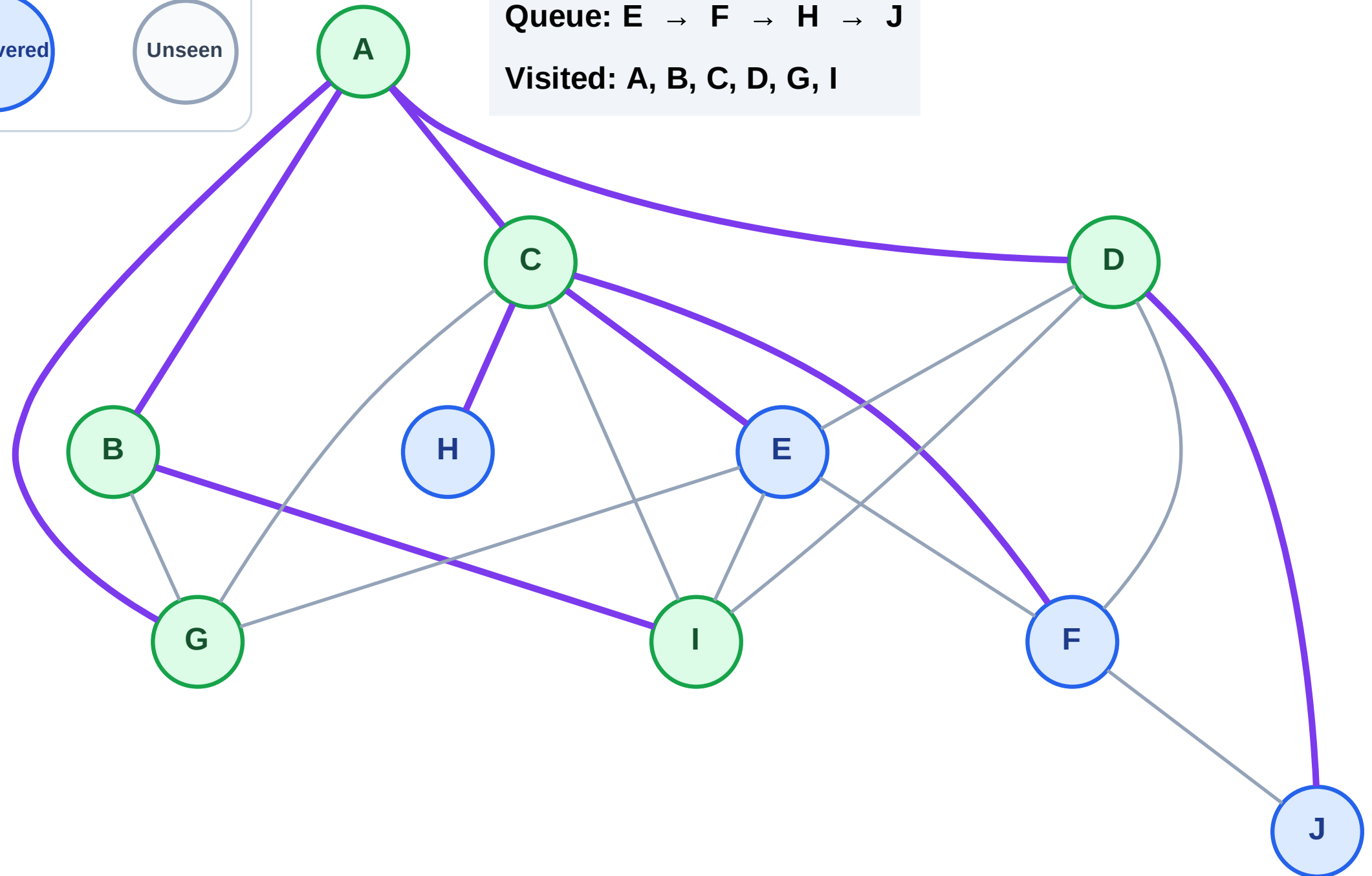
Step 13: No new neighbors from I

Legend



Queue: E → F → H → J

Visited: A, B, C, D, G, I



BFS Traversal

Step 14: Process node E

Legend

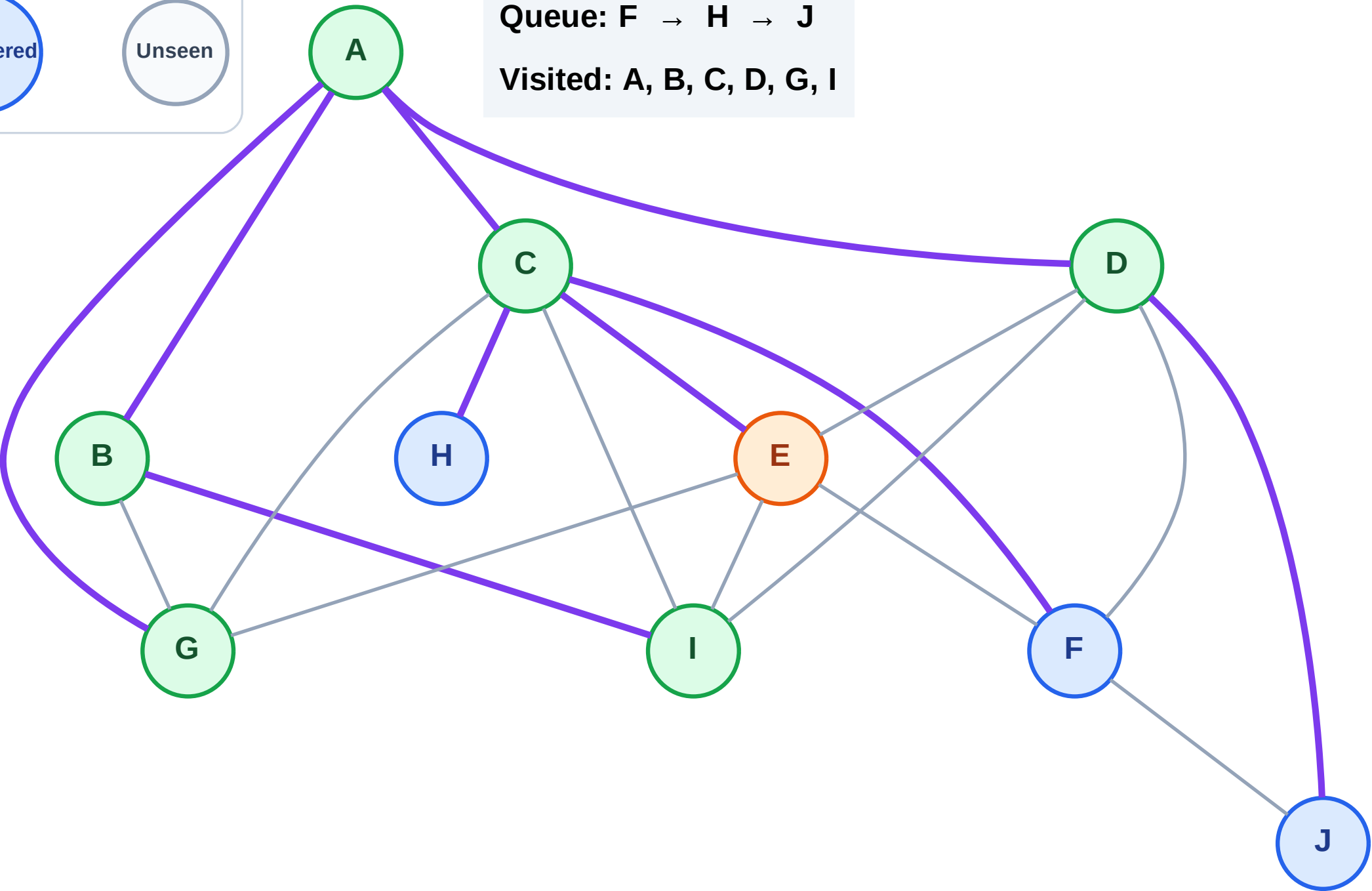
Complete

Processing

Discovered

Unseen

Queue: F → H → J
Visited: A, B, C, D, G, I



BFS Traversal

Step 15: No new neighbors from E

Legend

Complete

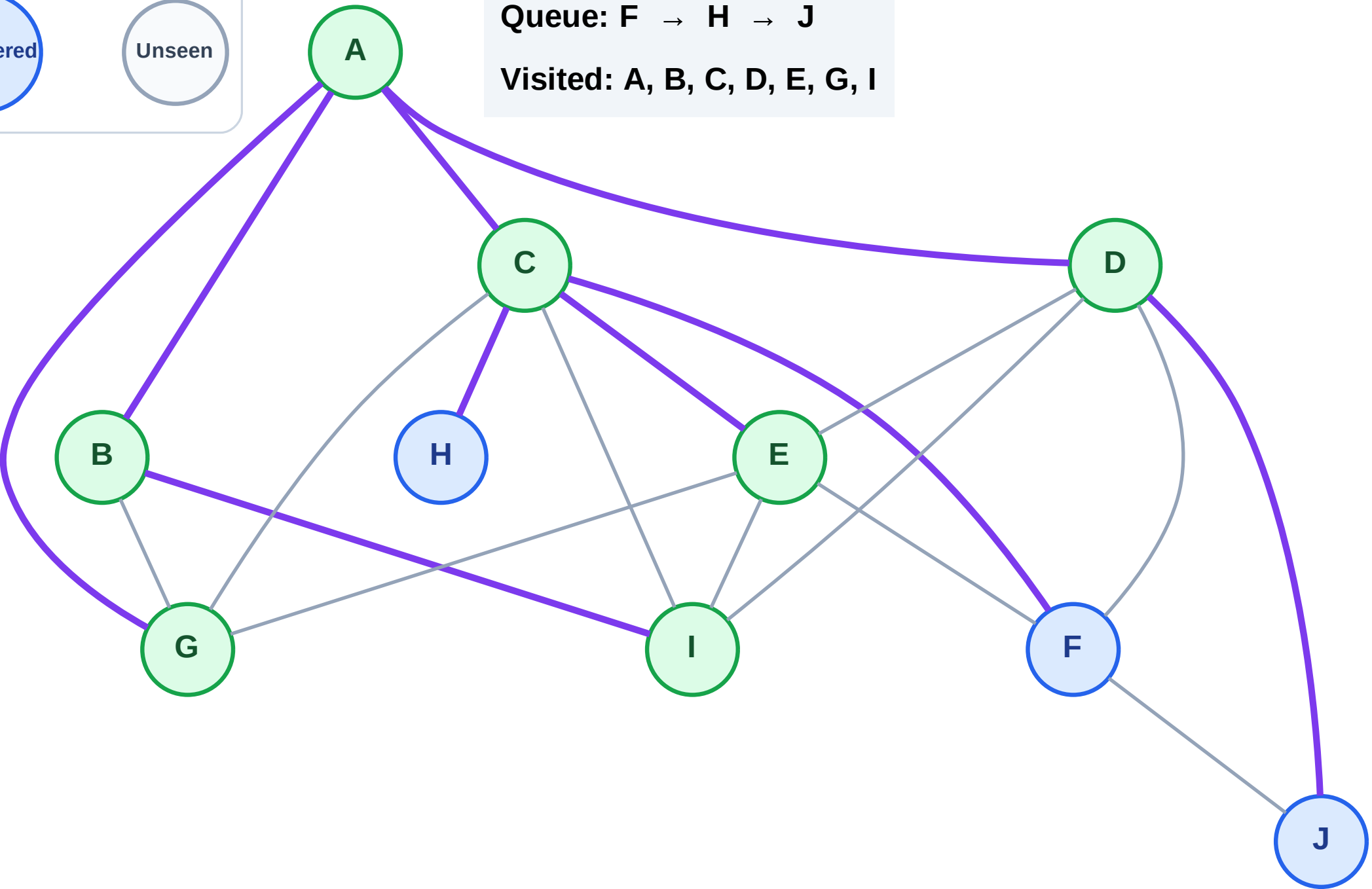
Processing

Discovered

Unseen

Queue: F → H → J

Visited: A, B, C, D, E, G, I



BFS Traversal

Step 16: Process node F

Legend

Complete

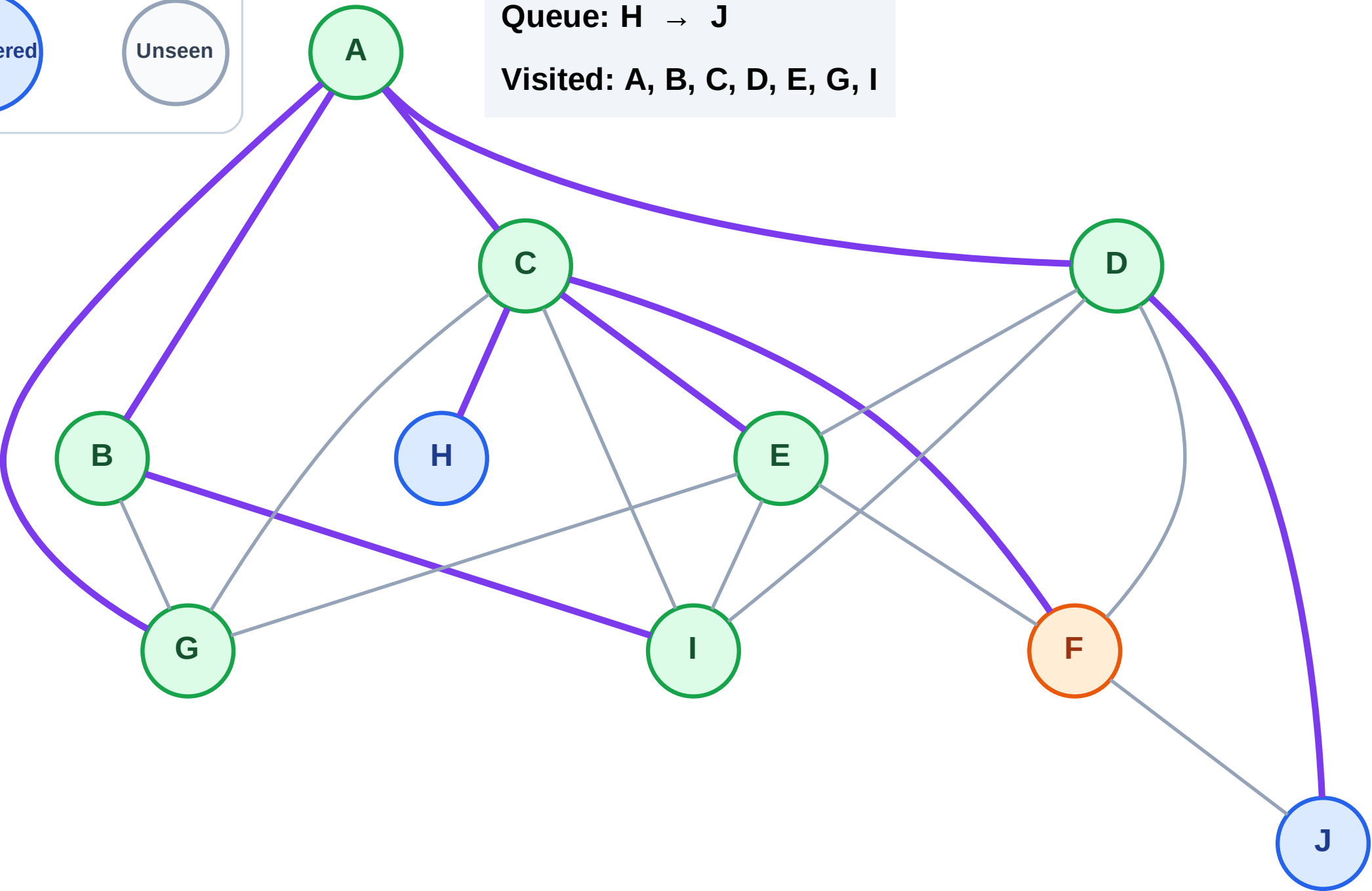
Processing

Discovered

Unseen

Queue: H → J

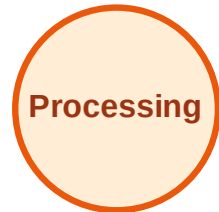
Visited: A, B, C, D, E, G, I



BFS Traversal

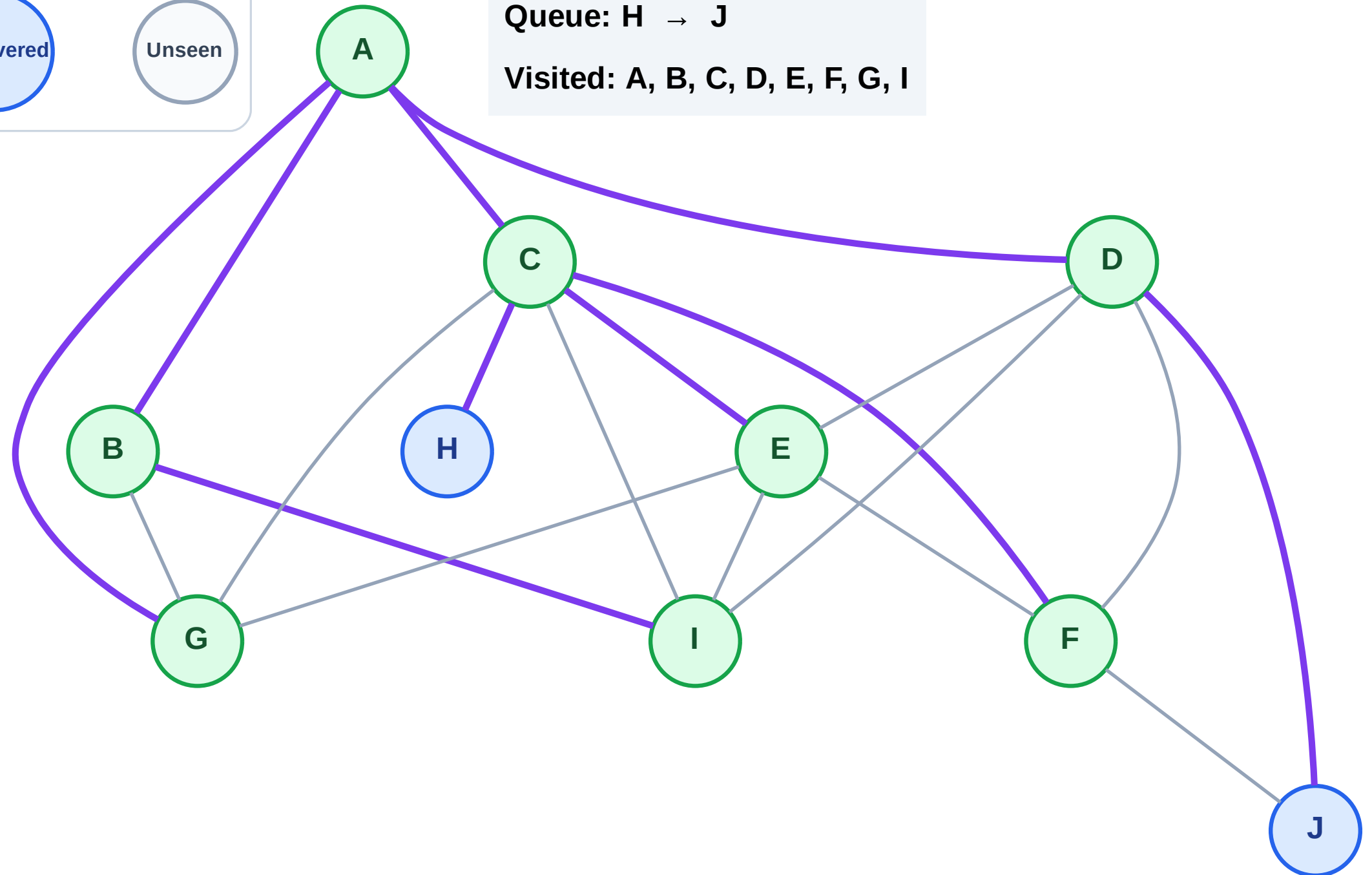
Step 17: No new neighbors from F

Legend



Queue: H → J

Visited: A, B, C, D, E, F, G, I



BFS Traversal

Step 18: Process node H

Legend

Complete

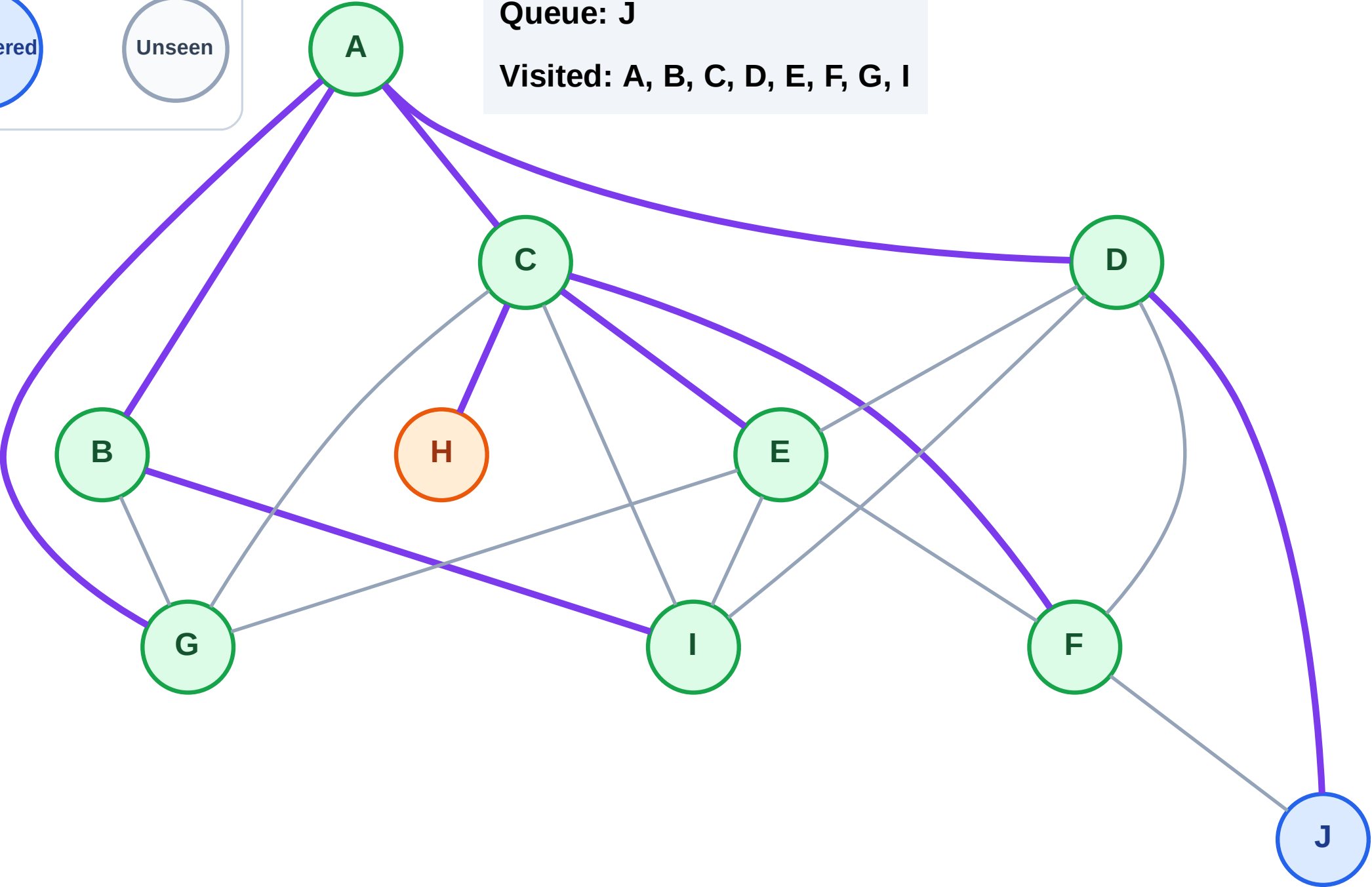
Processing

Discovered

Unseen

Queue: J

Visited: A, B, C, D, E, F, G, I



BFS Traversal

Step 19: No new neighbors from H

Legend

Complete

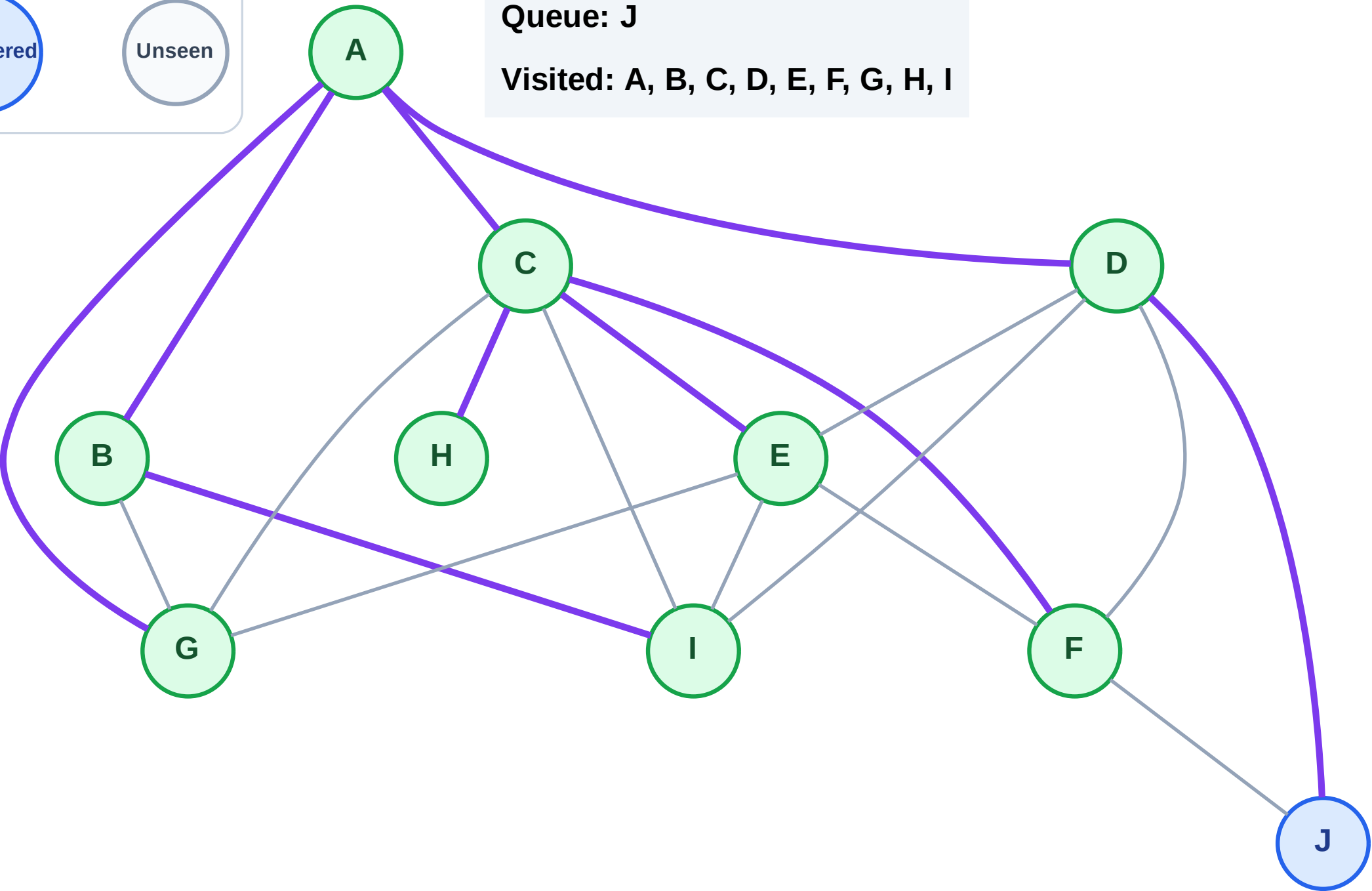
Processing

Discovered

Unseen

Queue: J

Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

Step 20: Process node J

Legend

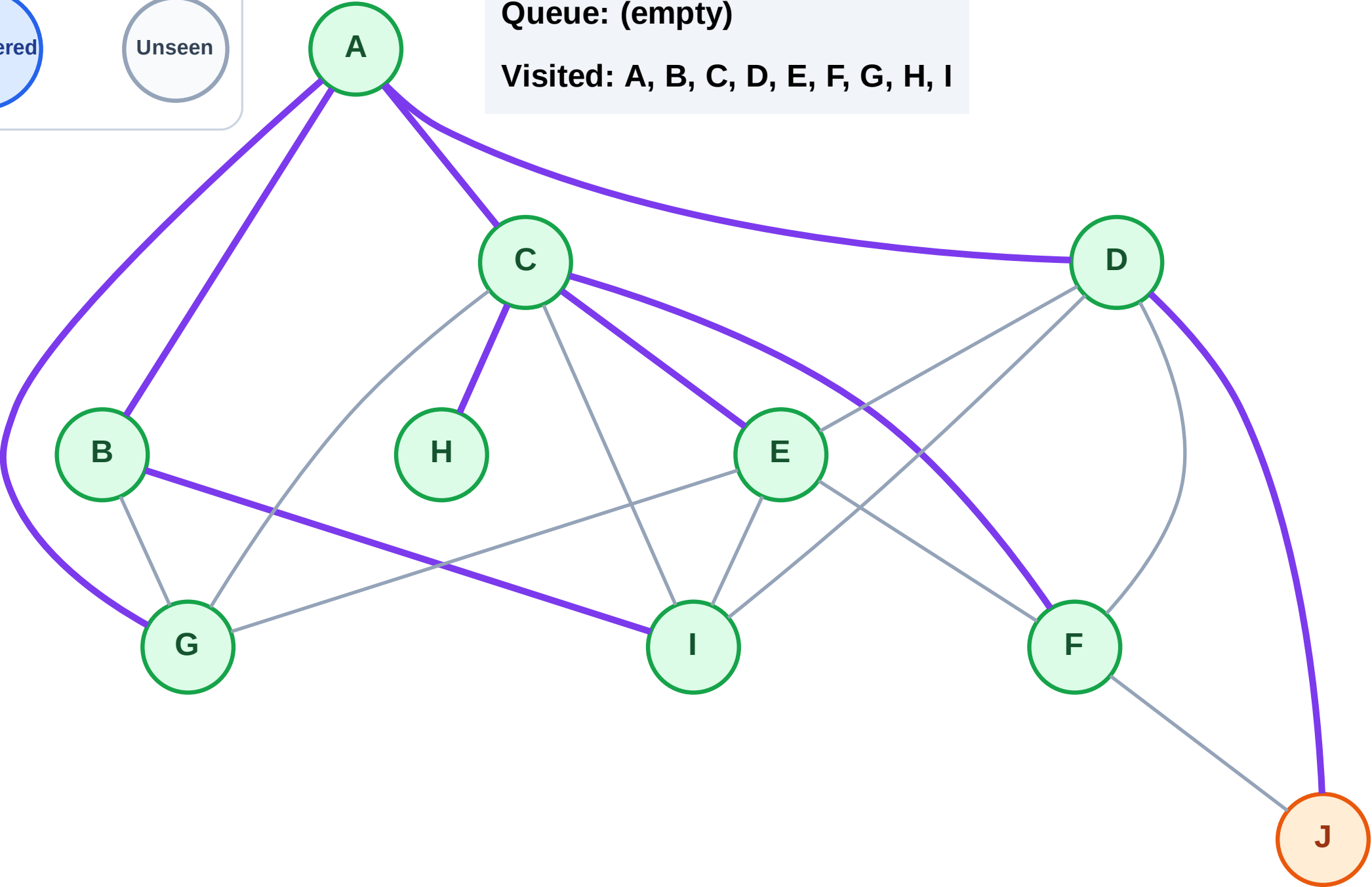
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

Step 21: No new neighbors from J

Legend

Complete

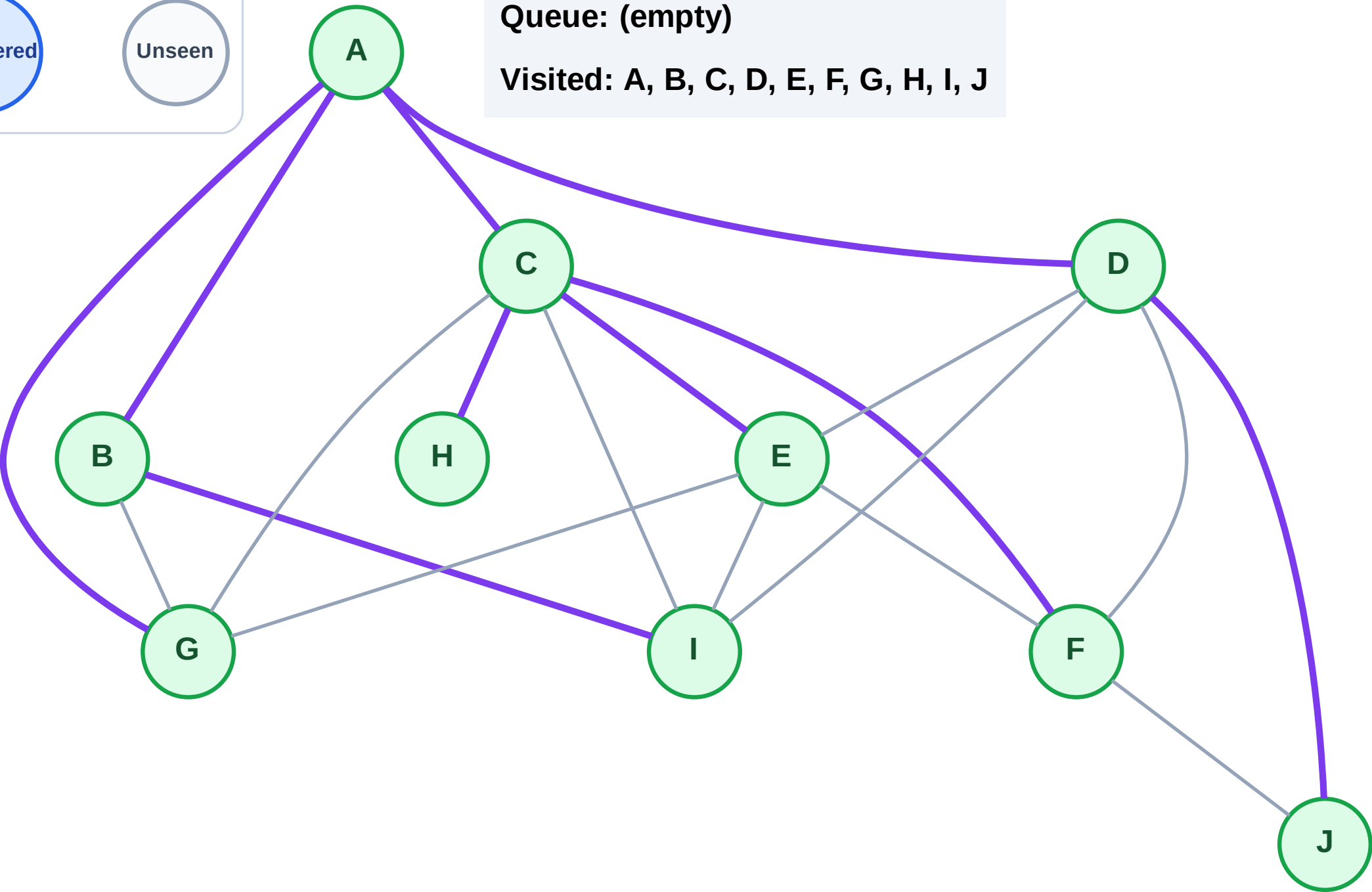
Processing

Discovered

Unseen

Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

Step 1: Start at A

Legend

Complete

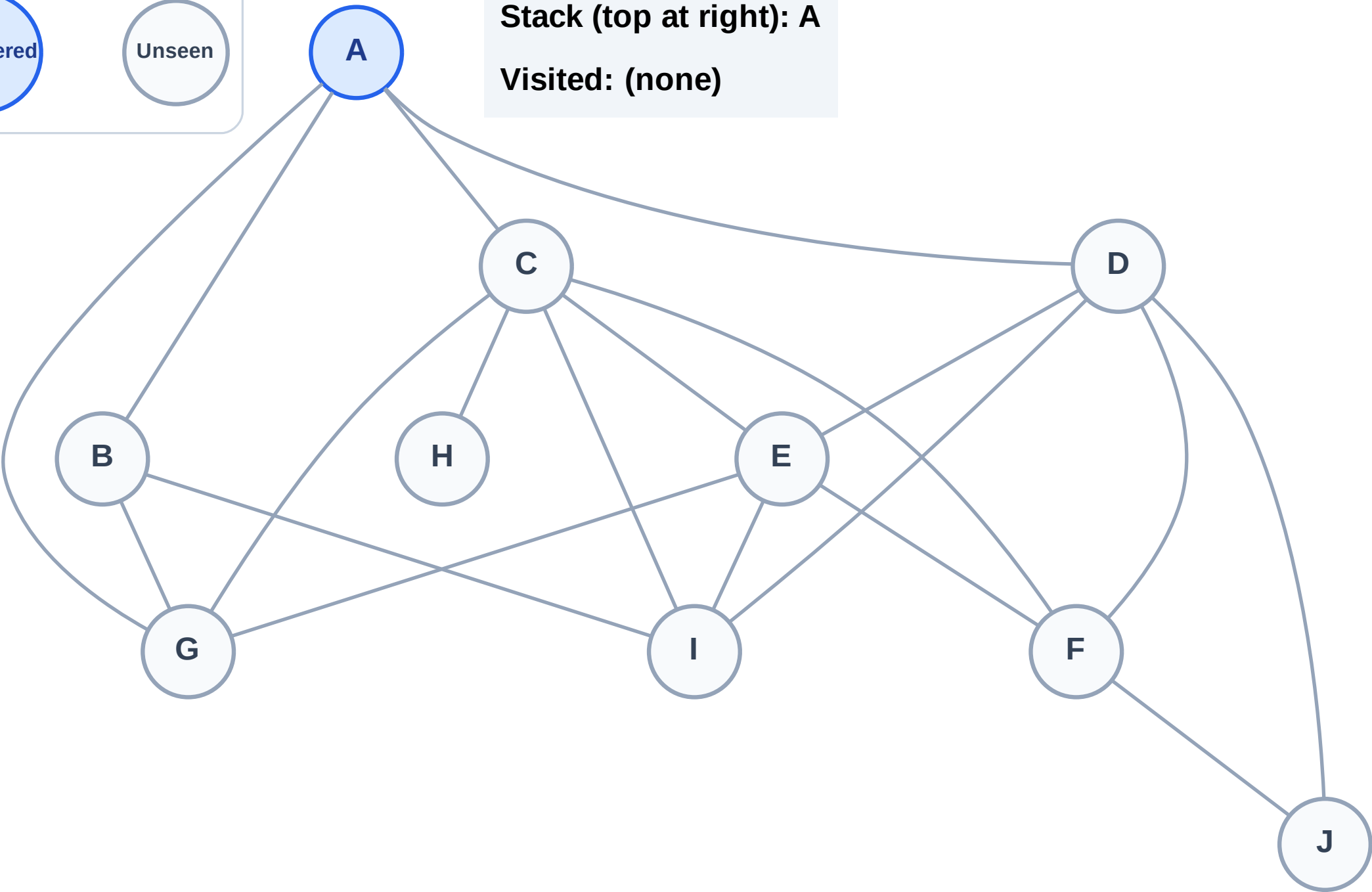
Processing

Discovered

Unseen

Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

Complete

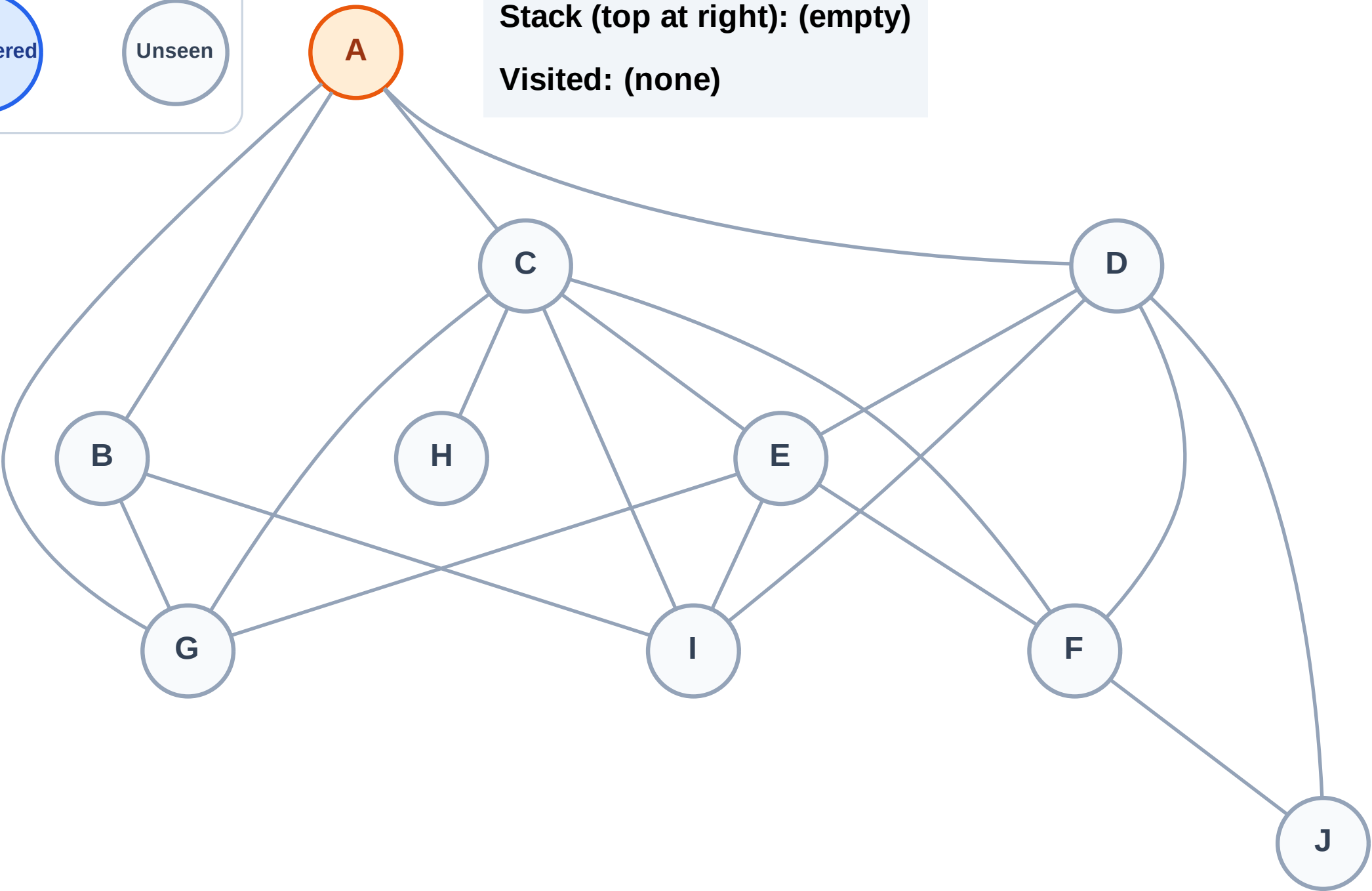
Processing

Discovered

Unseen

Stack (top at right): (empty)

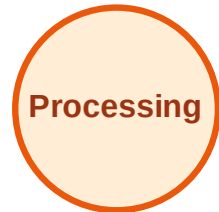
Visited: (none)



DFS Traversal

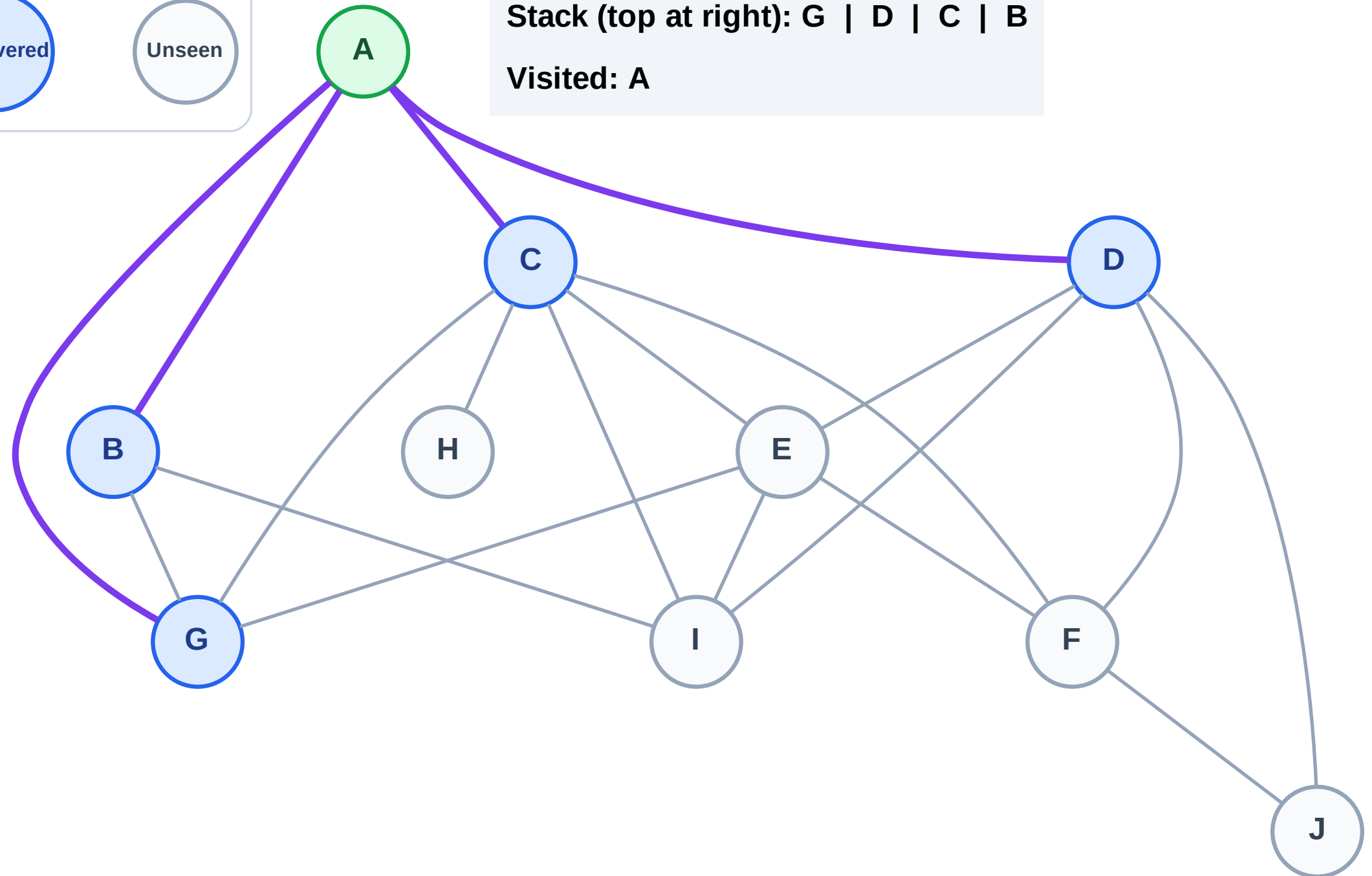
Step 3: Discover G, D, C, B from A

Legend



Stack (top at right): G | D | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

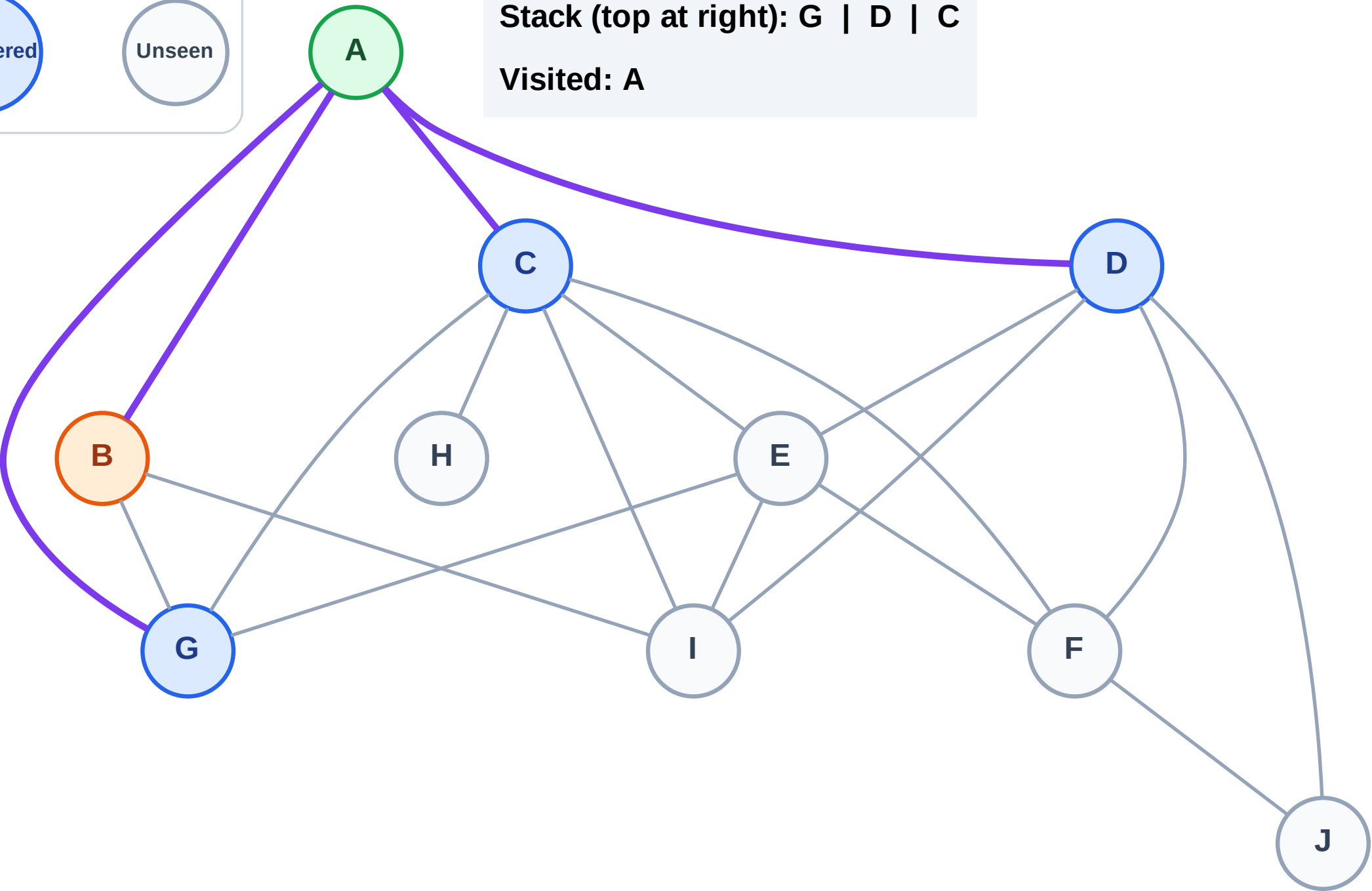
Processing

Discovered

Unseen

Stack (top at right): G | D | C

Visited: A



DFS Traversal

Step 5: Discover I from B

Legend

Complete

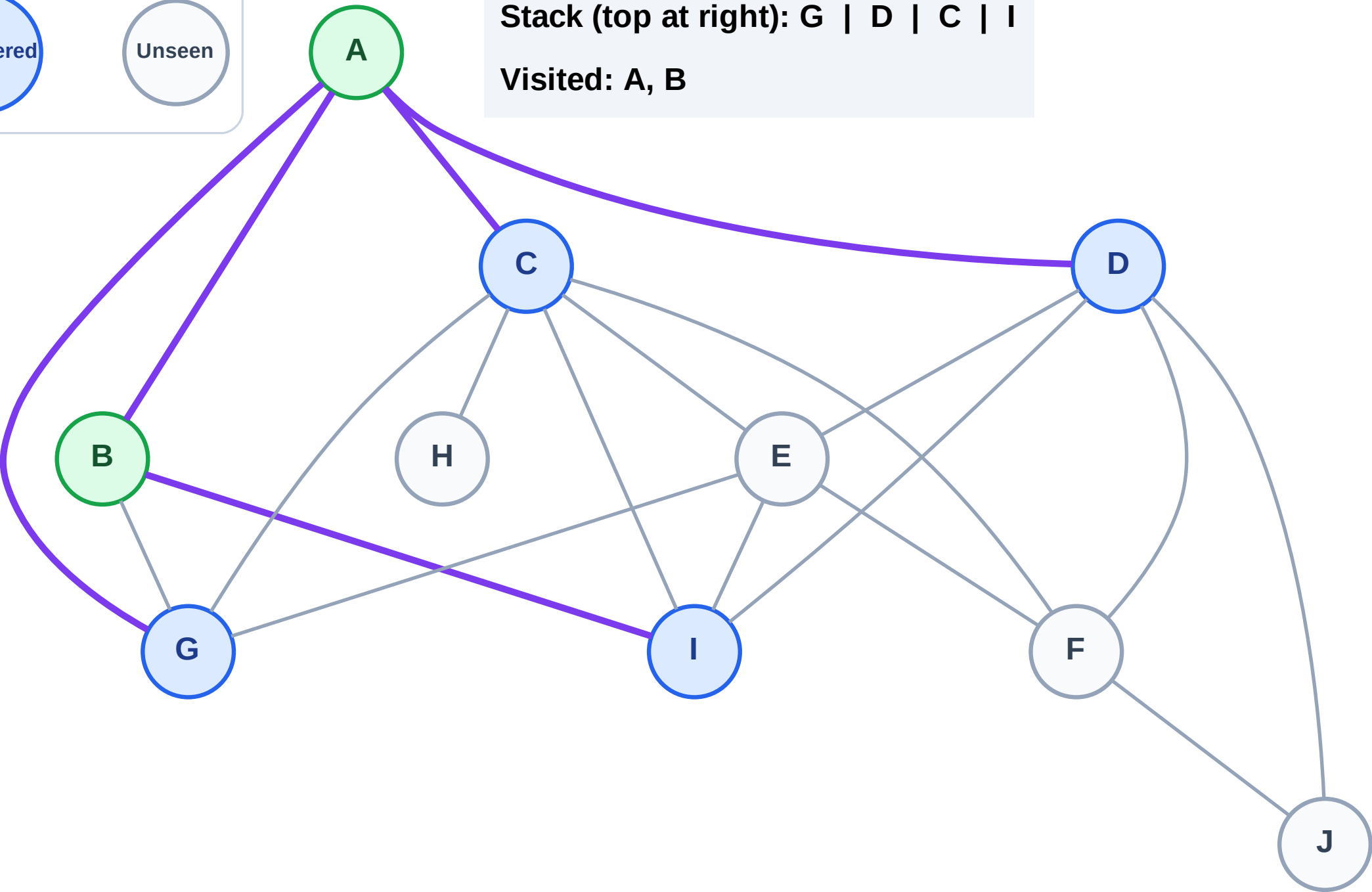
Processing

Discovered

Unseen

Stack (top at right): G | D | C | I

Visited: A, B



DFS Traversal

Step 6: Process node I

Legend

Complete

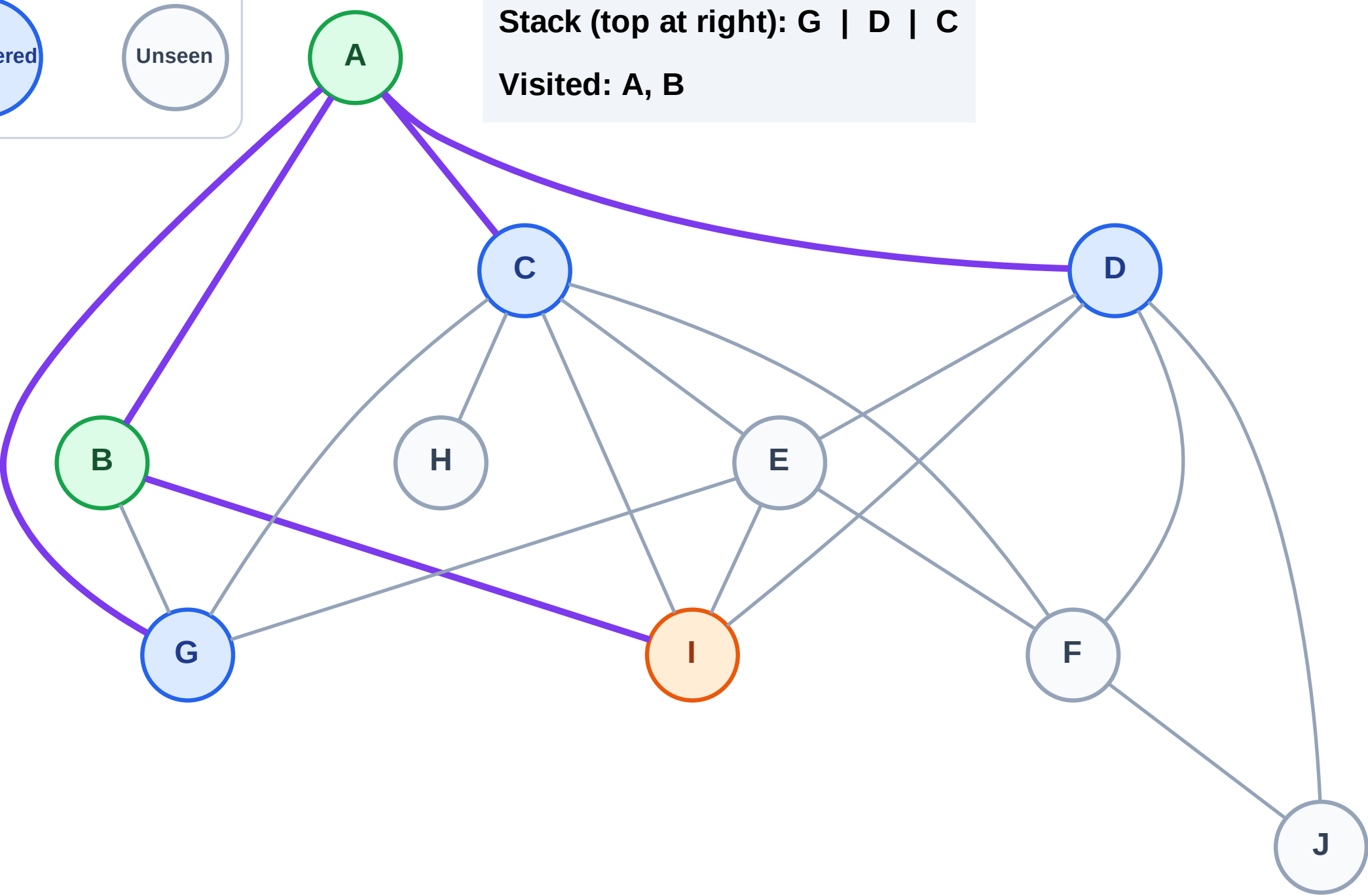
Processing

Discovered

Unseen

Stack (top at right): G | D | C

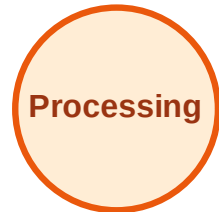
Visited: A, B



DFS Traversal

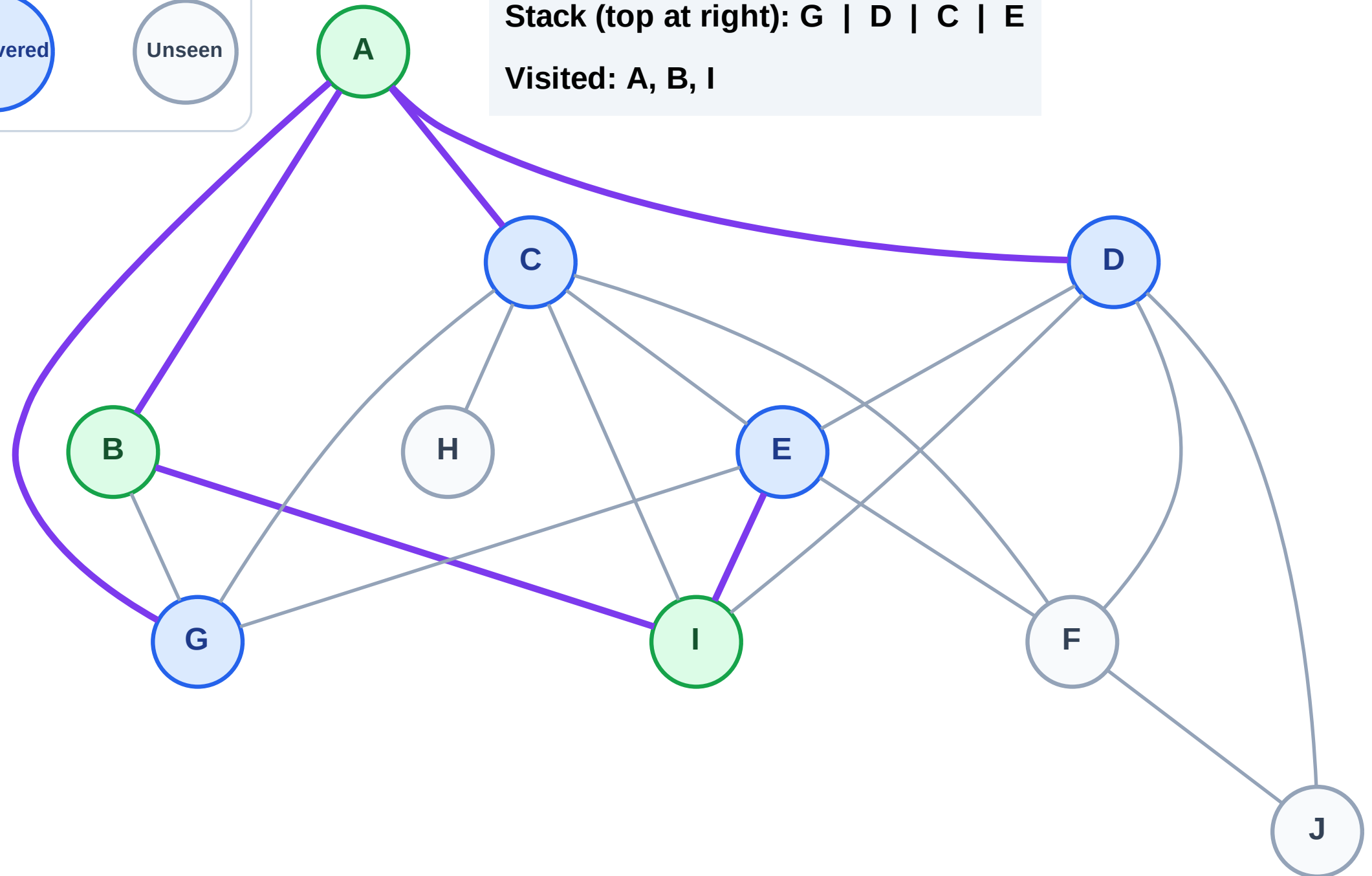
Step 7: Discover E from I

Legend



Stack (top at right): G | D | C | E

Visited: A, B, I



DFS Traversal

Step 8: Process node E

Legend

Complete

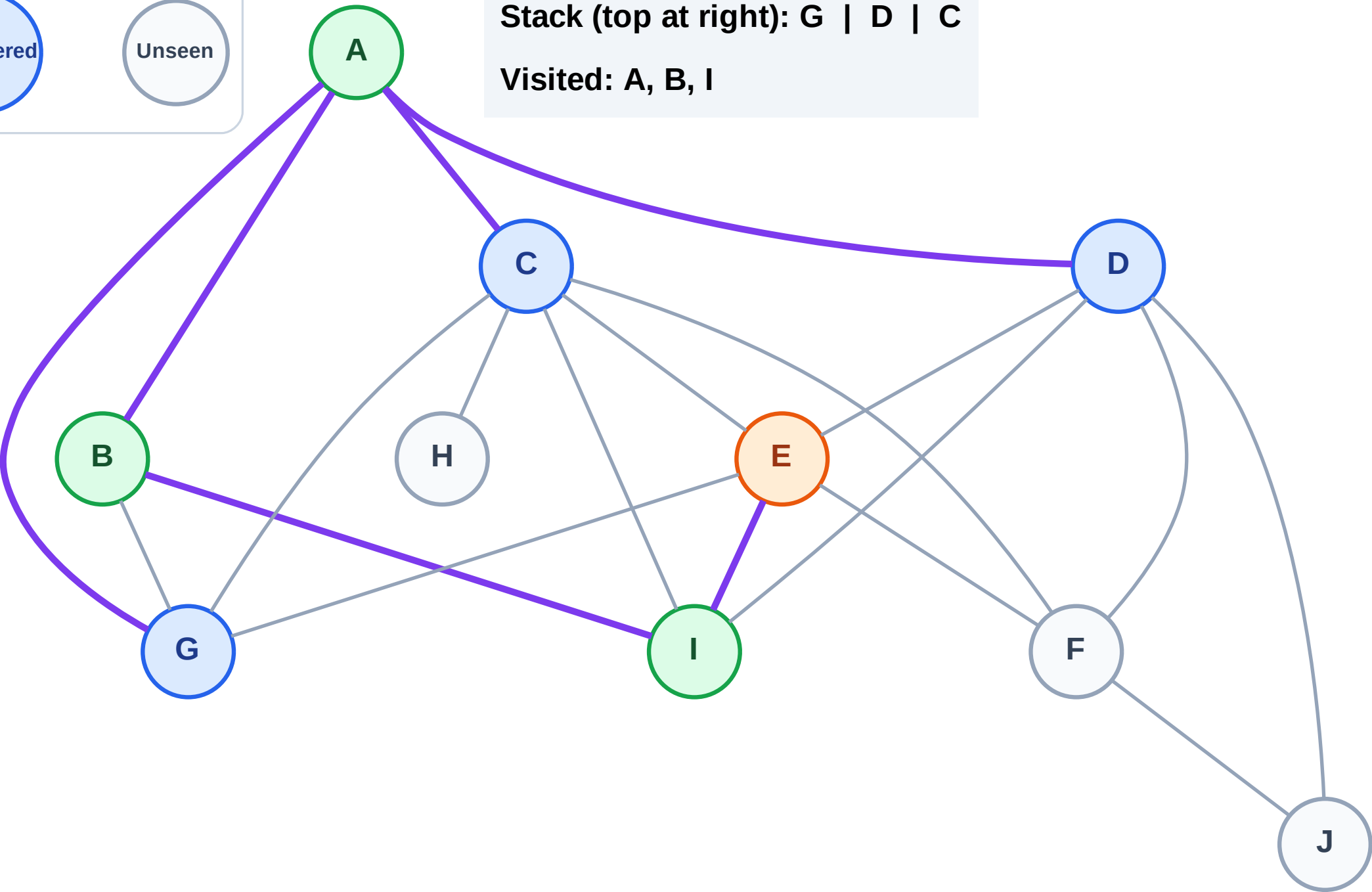
Processing

Discovered

Unseen

Stack (top at right): G | D | C

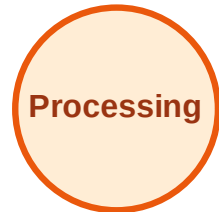
Visited: A, B, I



DFS Traversal

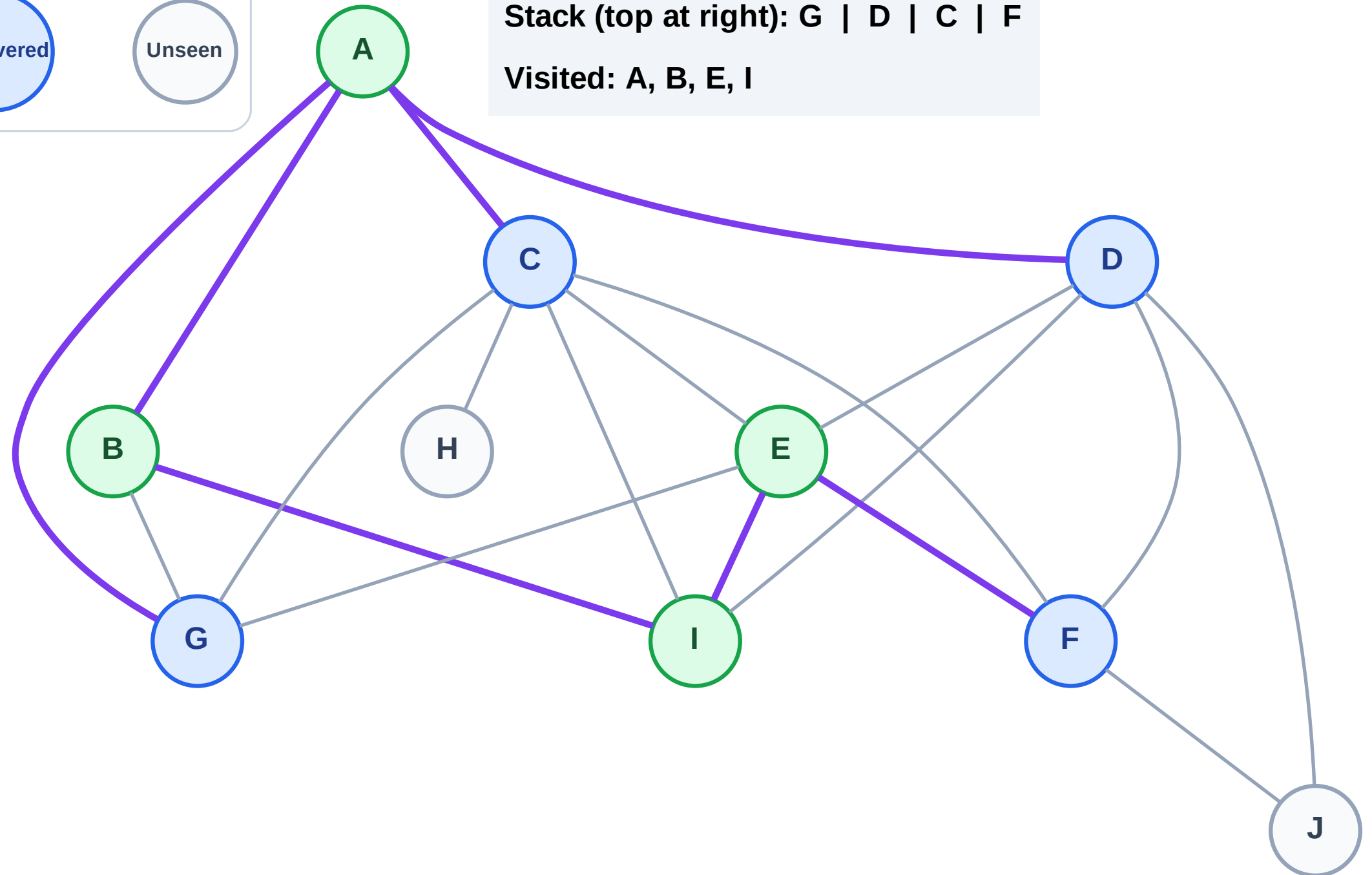
Step 9: Discover F from E

Legend



Stack (top at right): G | D | C | F

Visited: A, B, E, I



DFS Traversal

Step 10: Process node F

Legend

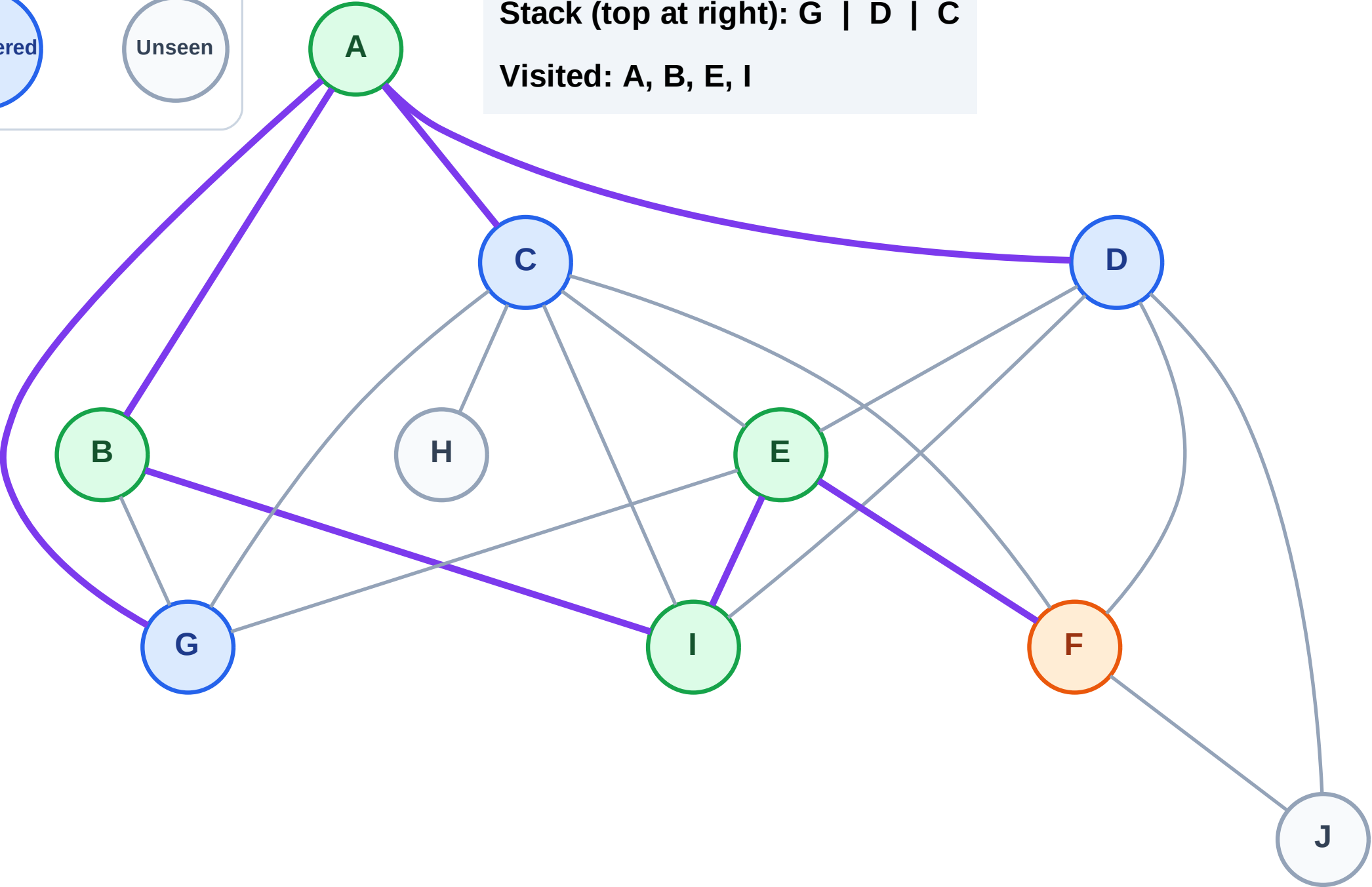
Complete

Processing

Discovered

Unseen

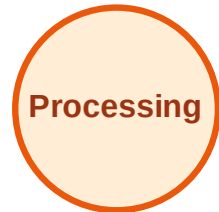
Stack (top at right): G | D | C
Visited: A, B, E, I



DFS Traversal

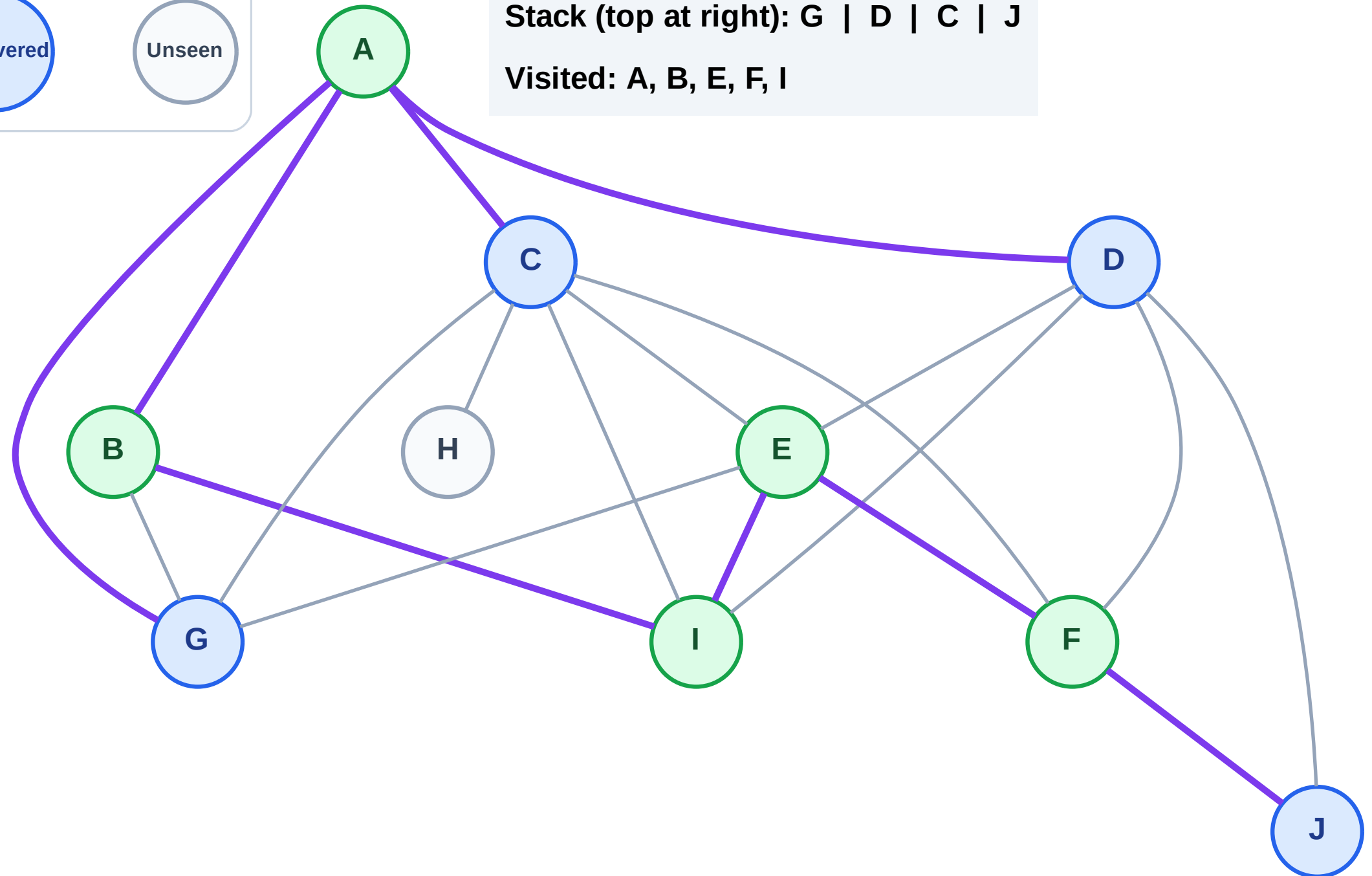
Step 11: Discover J from F

Legend



Stack (top at right): G | D | C | J

Visited: A, B, E, F, I



DFS Traversal

Step 12: Process node J

Legend

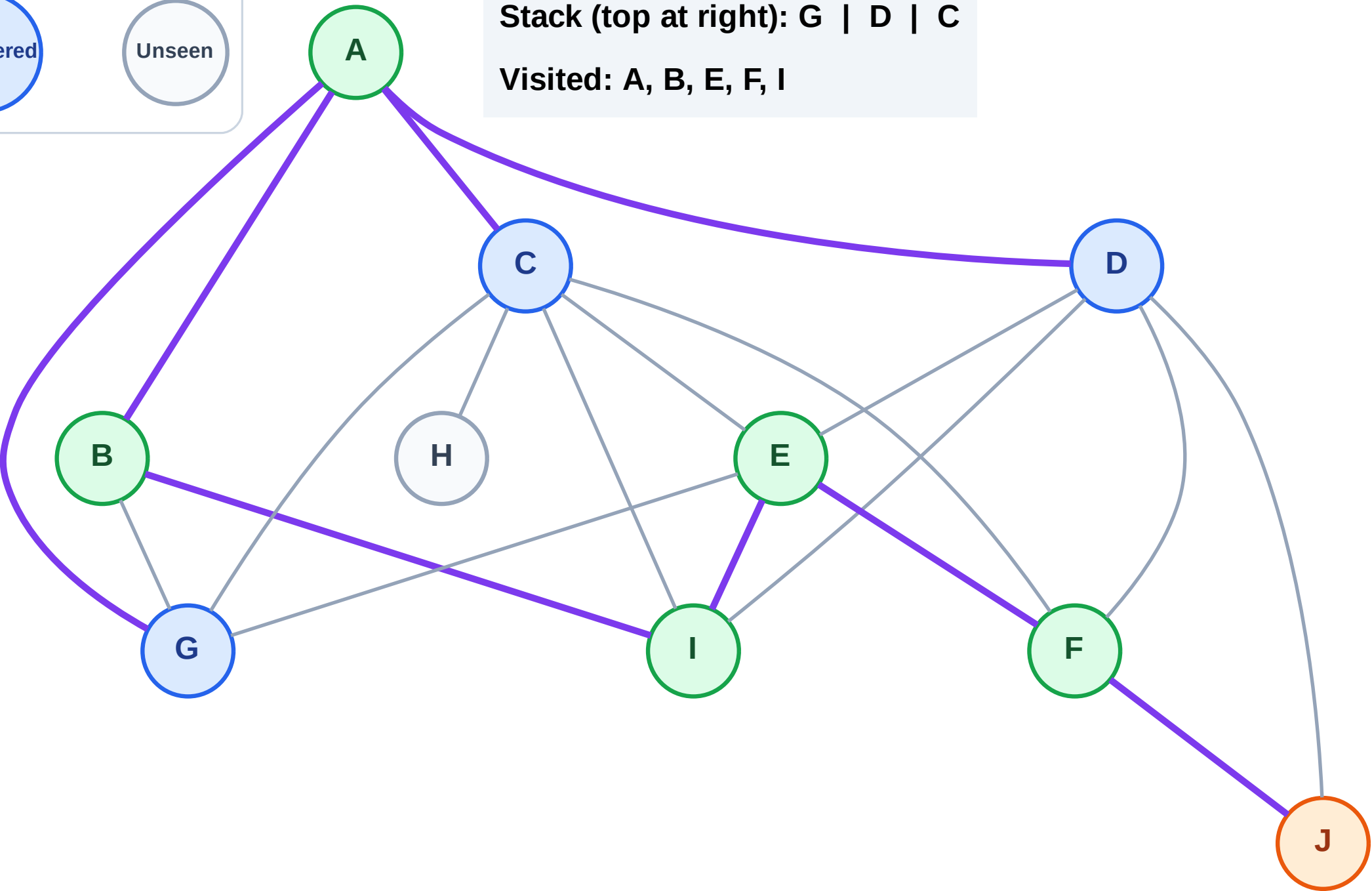
Complete

Processing

Discovered

Unseen

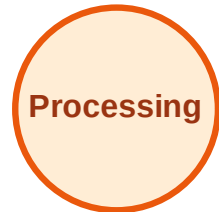
Stack (top at right): G | D | C
Visited: A, B, E, F, I



DFS Traversal

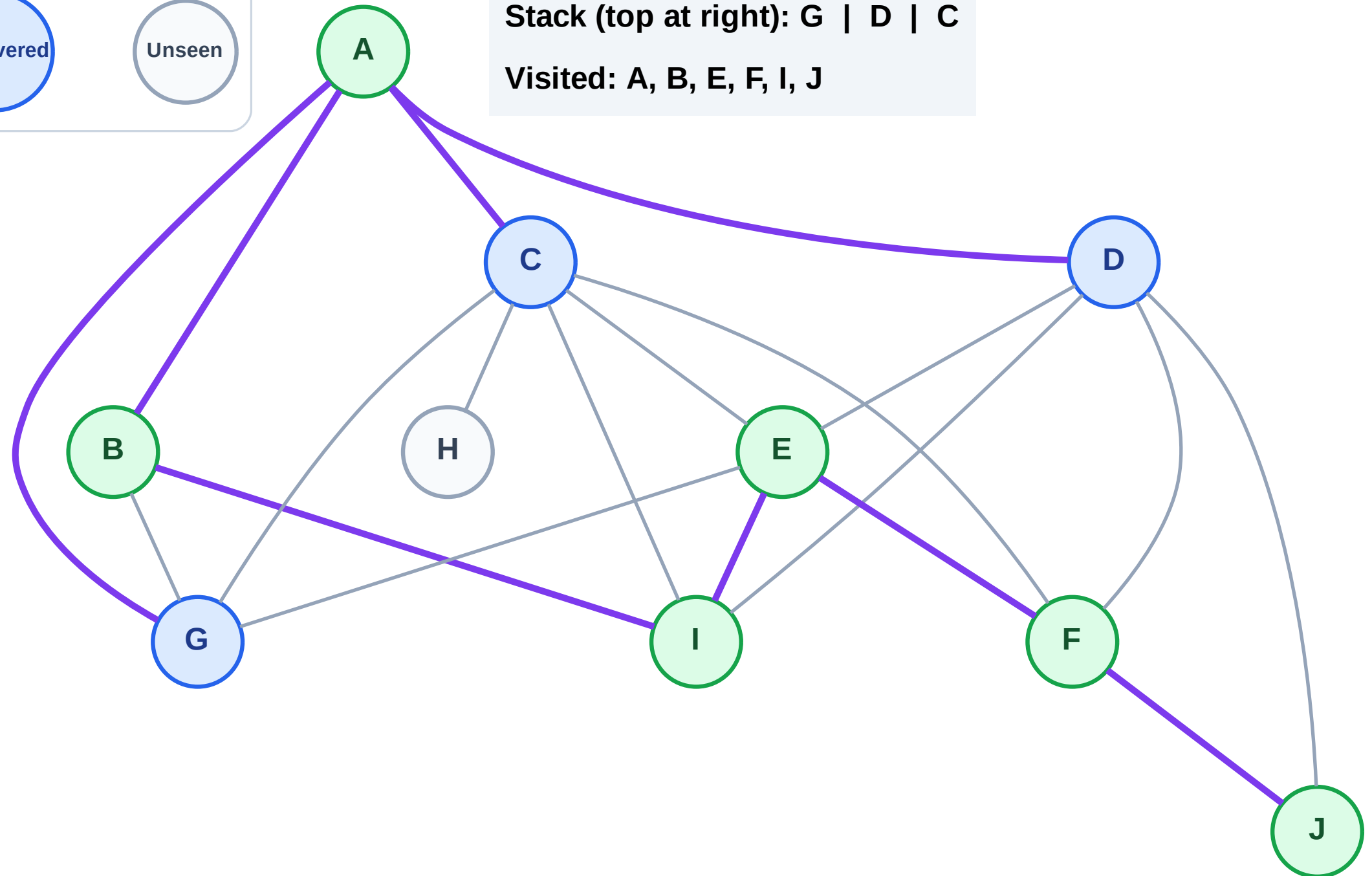
Step 13: No new neighbors from J

Legend



Stack (top at right): G | D | C

Visited: A, B, E, F, I, J



DFS Traversal

Step 14: Process node C

Legend

Complete

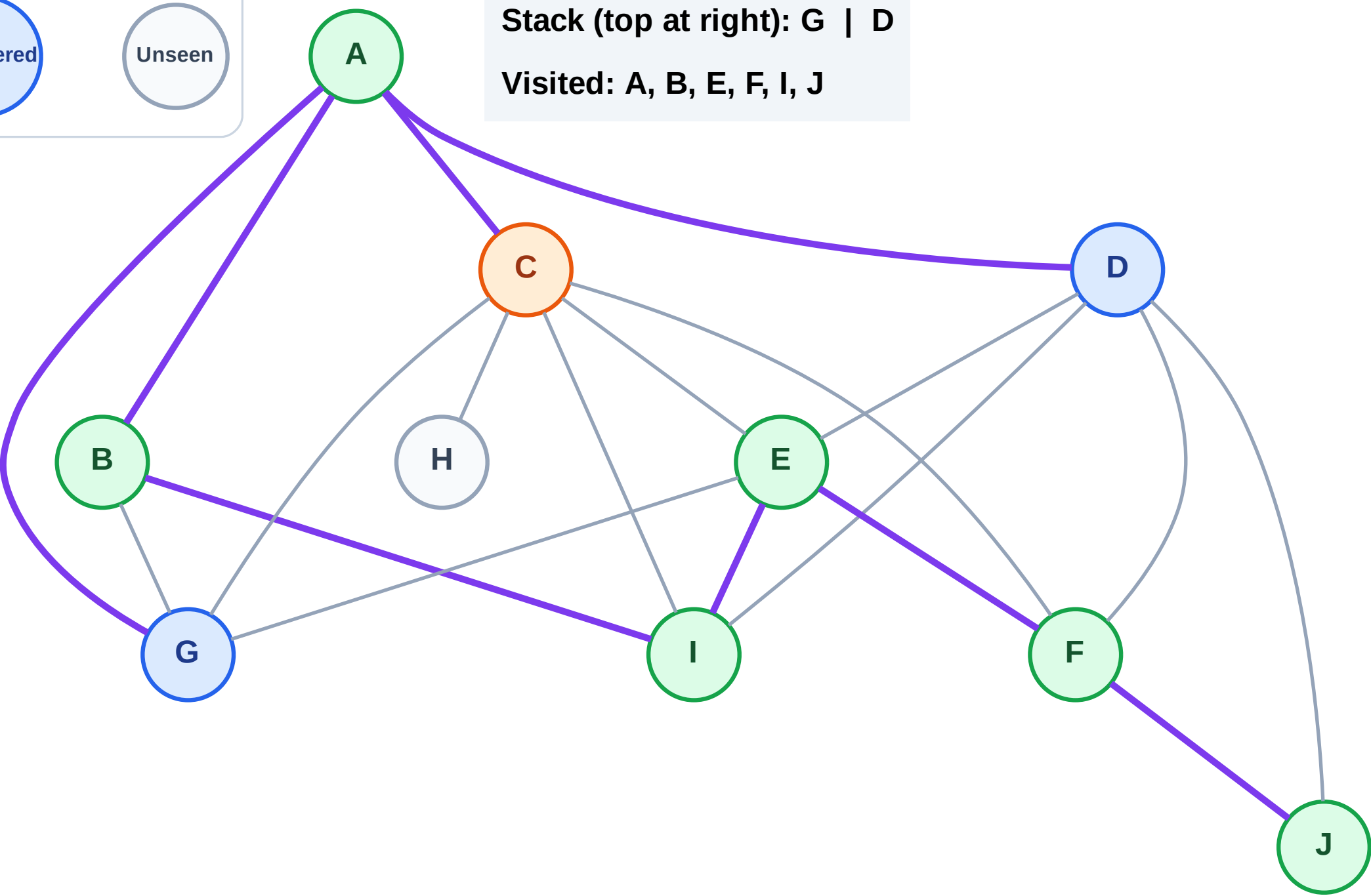
Processing

Discovered

Unseen

Stack (top at right): G | D

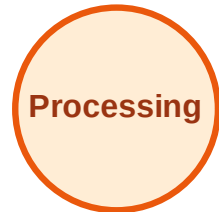
Visited: A, B, E, F, I, J



DFS Traversal

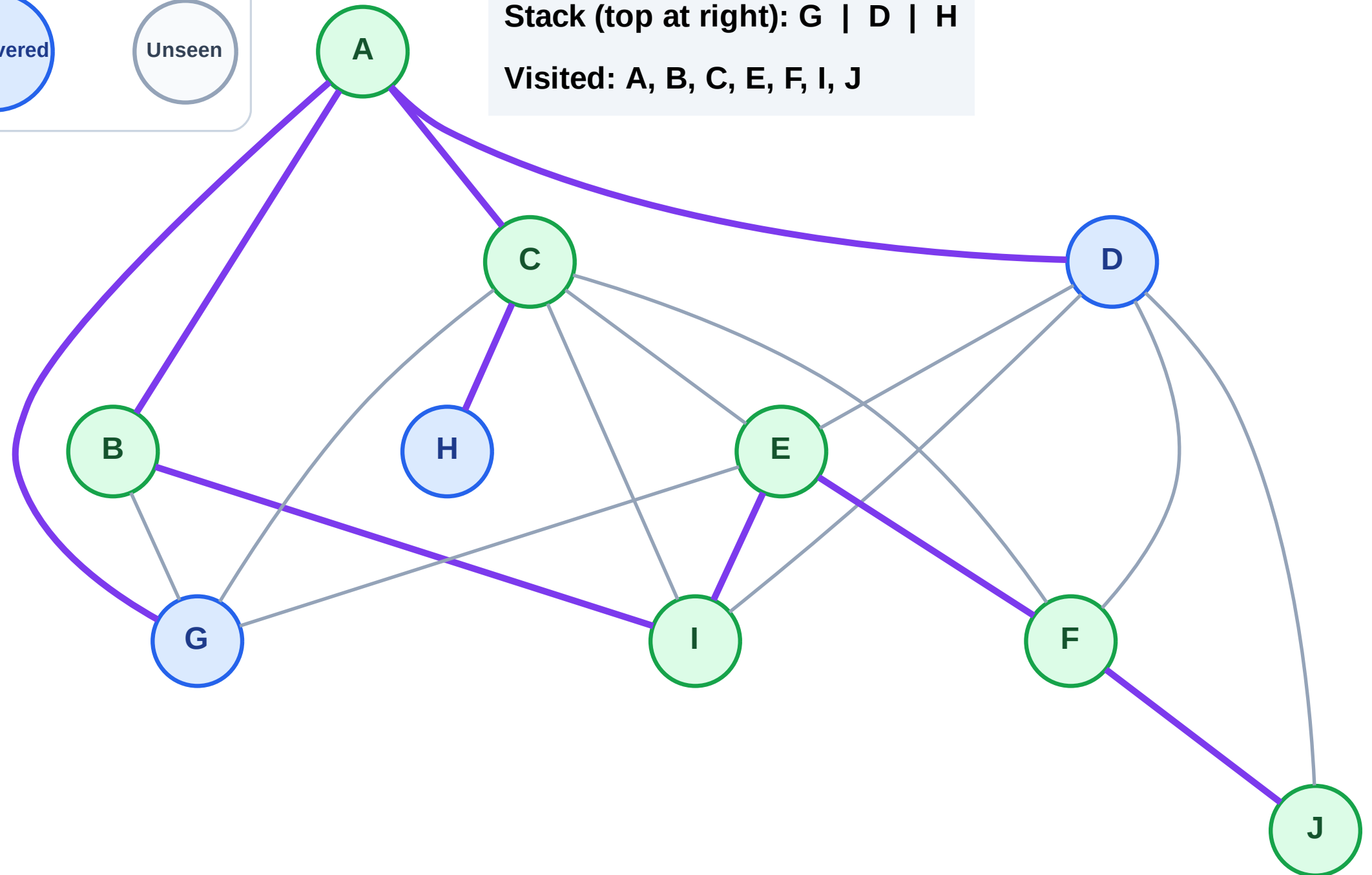
Step 15: Discover H from C

Legend



Stack (top at right): G | D | H

Visited: A, B, C, E, F, I, J



DFS Traversal

Step 16: Process node H

Legend

Complete

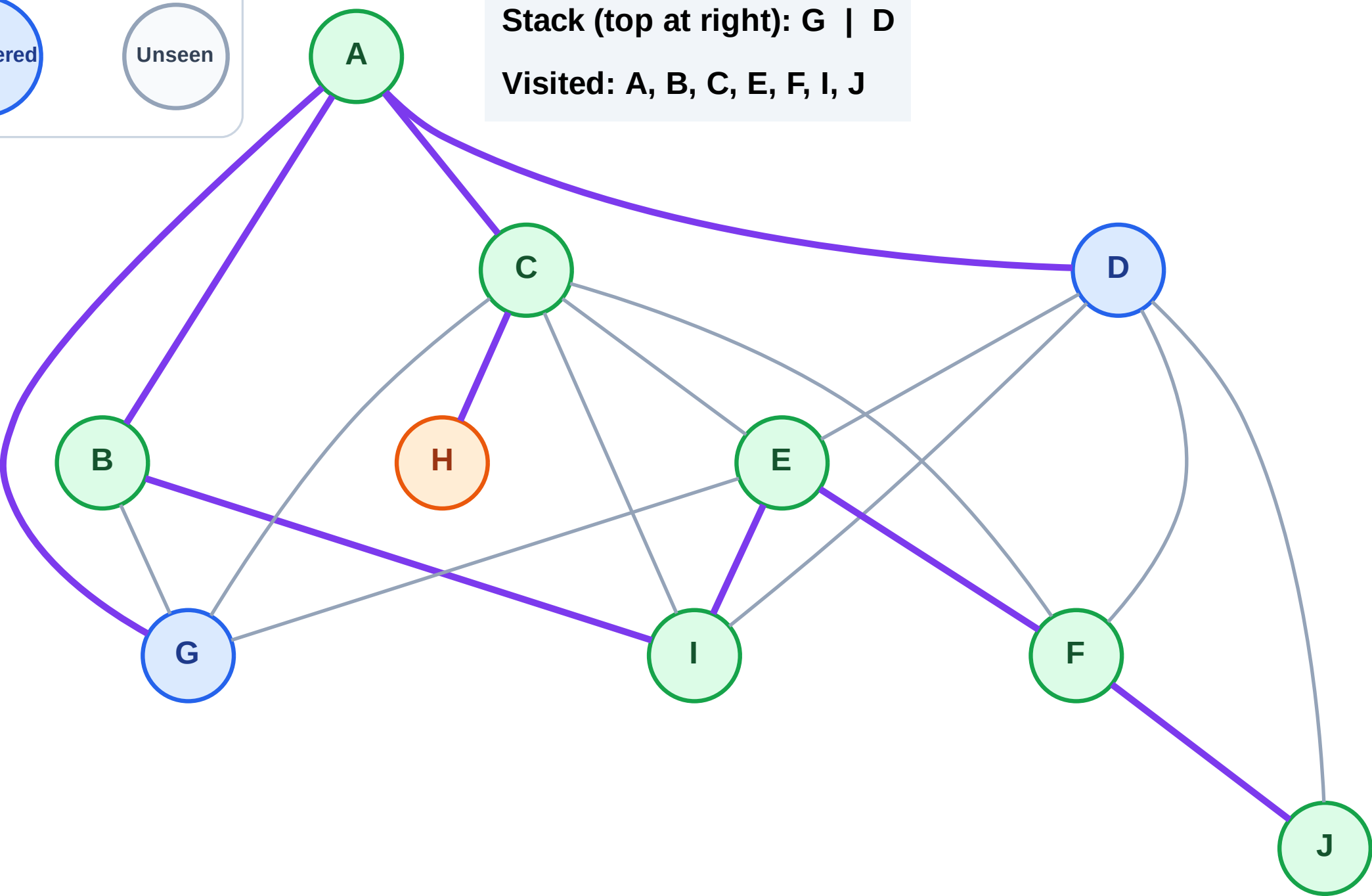
Processing

Discovered

Unseen

Stack (top at right): G | D

Visited: A, B, C, E, F, I, J



DFS Traversal

Step 17: No new neighbors from H

Legend

Complete

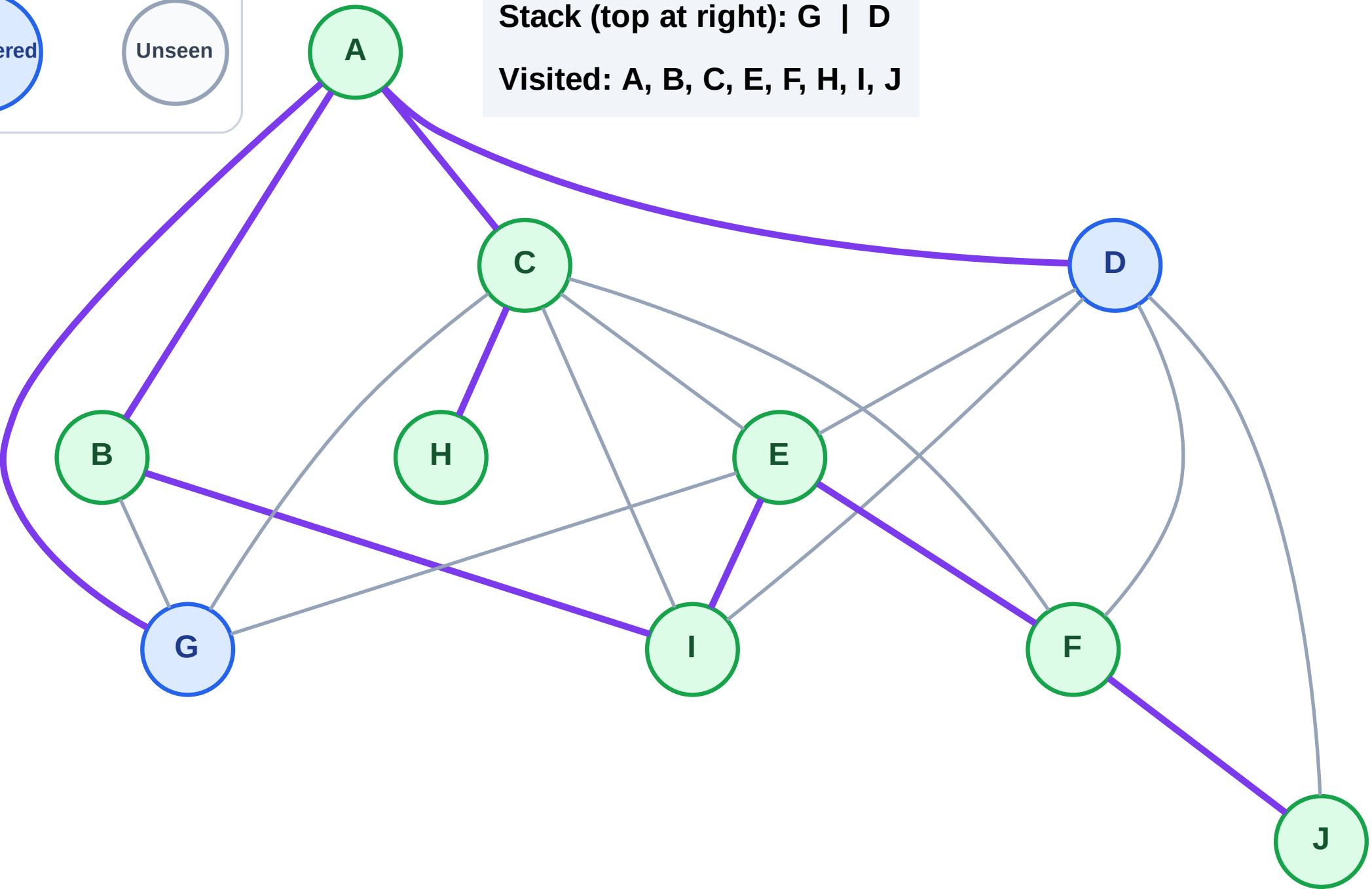
Processing

Discovered

Unseen

Stack (top at right): G | D

Visited: A, B, C, E, F, H, I, J



DFS Traversal

Step 18: Process node D

Legend

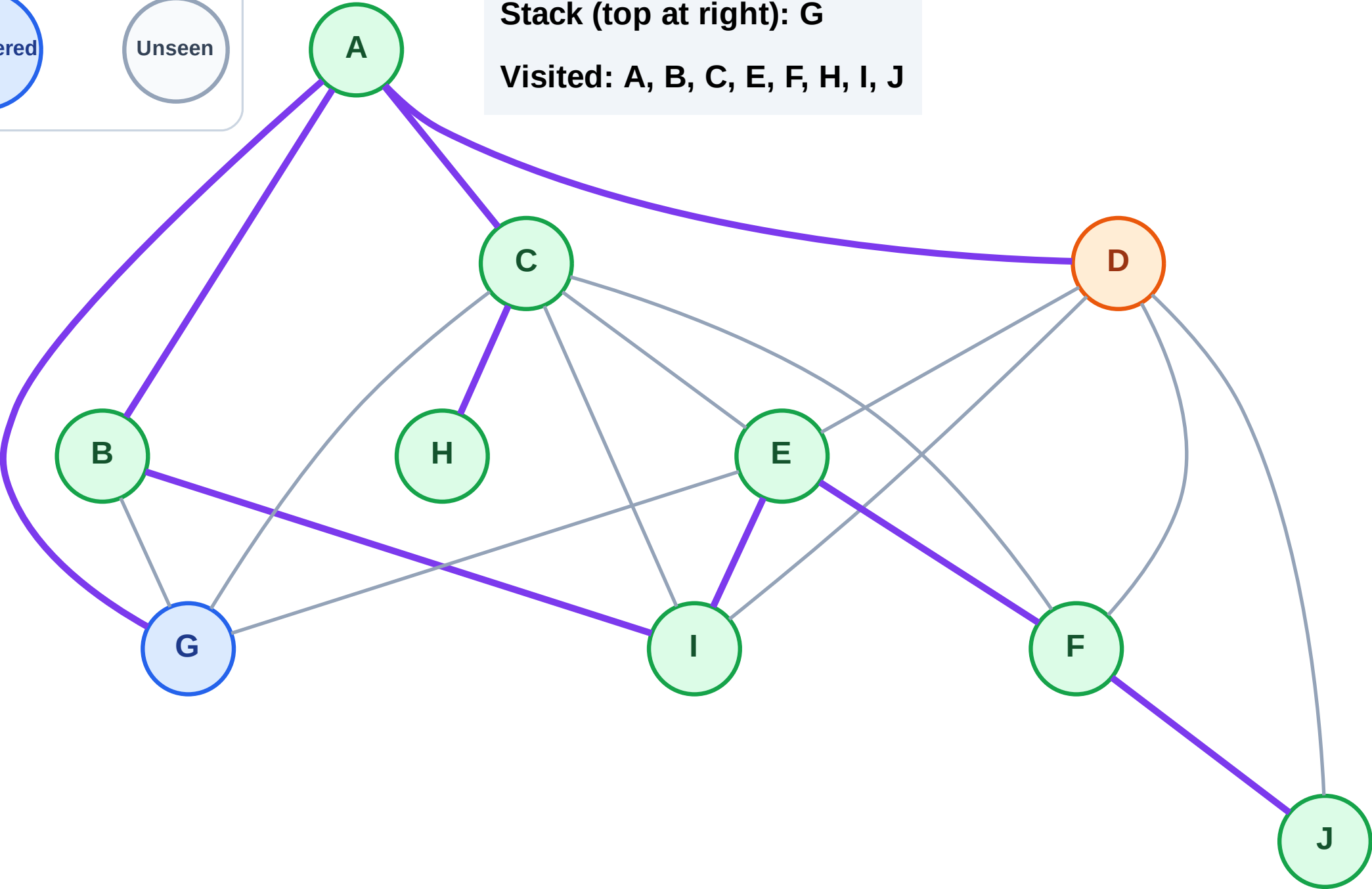
Complete

Processing

Discovered

Unseen

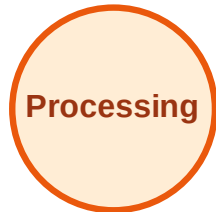
Stack (top at right): G
Visited: A, B, C, E, F, H, I, J



DFS Traversal

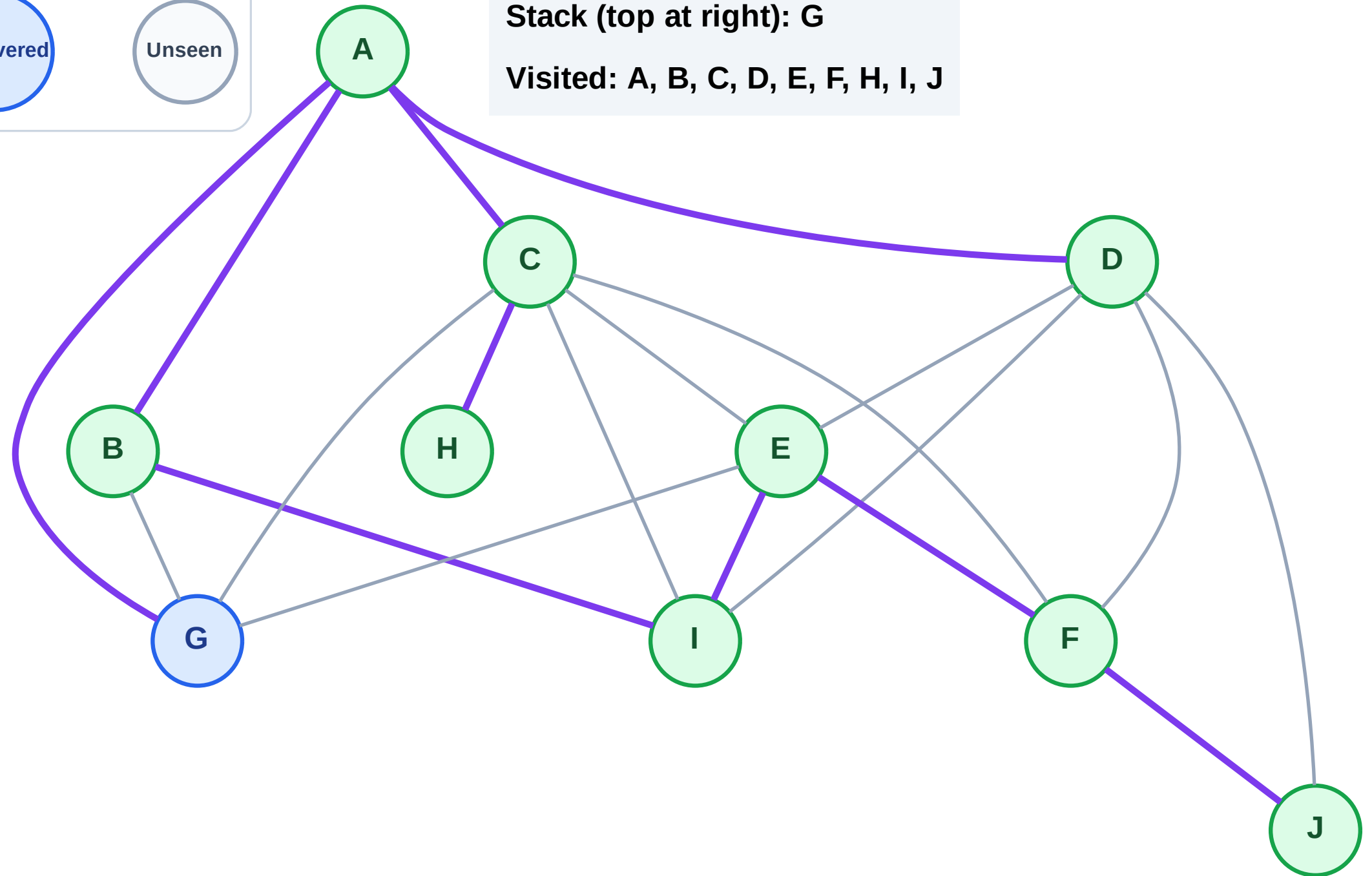
Step 19: No new neighbors from D

Legend



Stack (top at right): G

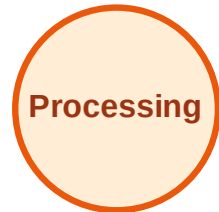
Visited: A, B, C, D, E, F, H, I, J



DFS Traversal

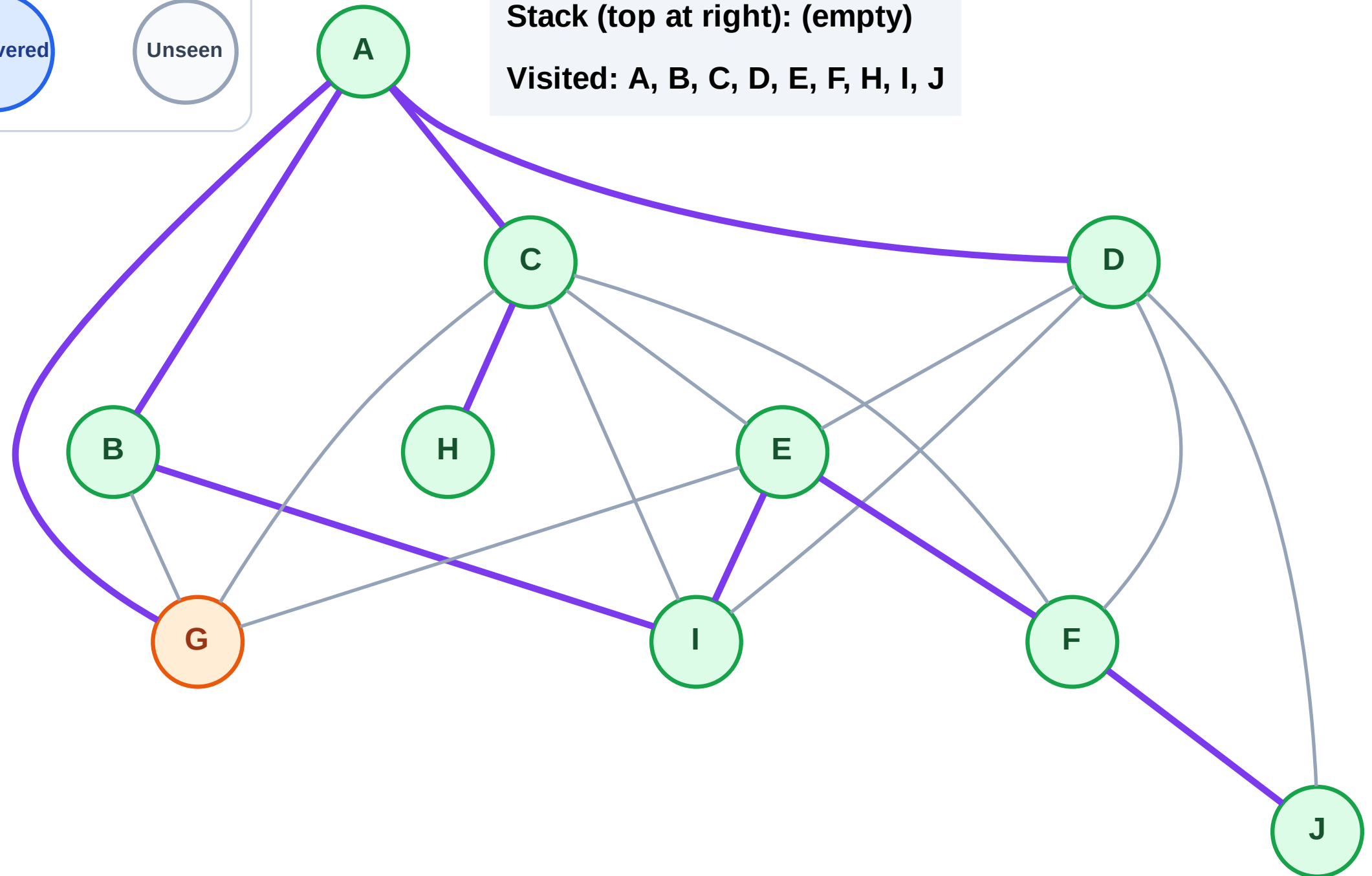
Step 20: Process node G

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, H, I, J



DFS Traversal

Step 21: No new neighbors from G

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

